What Did Ten Points of Mean Recall Buy? Separating Group-Level from Case-Level Gains in Scene Graph Generation

Supplementary Material

A. Contents of this supplementary material

Section letters below are this document's; numbered sections, equations, tables and figures cited without a letter are the main paper's. One naming difference: this supplementary keeps the long-form names for the two parts of a gain. The main paper's *group-level* part ΔP is called the *transported gain* here, and its *case-level* part ΔR the *within-group covariance gain*; the quantities are identical. Appendix B states the eleven formal results the paper uses and Appendix E proves them. The remaining sections give the worked four-point example (Appendix C), the influence-function derivation (Appendix F), the randomization algorithm (Appendix G), implementation, architectures, seeds and compute (Appendix H), the sensitivity bound evaluated on our own estimates (Appendix I), the conditions of the limit theorem (Appendix J), the choice of grouping variable (Appendix K), a terminology table (Appendix L), the four-point example showing that the group-level part is not prior fit (Appendix M), and the full protocols and tables for the controlled predictors, the CLIP comparison, the sensitivity analyses and the two long-tailed studies (Appendix N), and the mean-recall split of the released checkpoint with the evaluator replay, the operating-point control, the within-cell AUC and the proper-score split in every configuration (Appendix O), the CLIP study over five seeds (Appendix P), and the scene-graph audit under 20 calibration halves, merged predicates, a path of label mixes, both AUC weightings, predicate-tier pairs, three recalibration families and a zero-shot CLIP model (Appendix Q).

The code, the results files behind every number, the scripts that regenerate each model's predictions, and a script that checks every number printed in the paper against the results files are uploaded as an anonymized archive alongside this document.

B. Formal results

The main paper states one result, Theorem 1, and uses eleven more. They are stated here in the order the paper relies on them; proofs are in Appendix E. Equation and section references to the main paper resolve to its numbering, since both are compiled as one document.

B.1. What the group-level part measures

Proposition 1 (Composition of the transported gain)**.** *Under the quadratic score, write $\bar{q}_m(c) = \mathbb{E}[q_m(\cdot \mid X) \mid C = c]$ for model m's mean forecast in cell c, $\mathrm{Var}(q_m \mid c)$ for the within-cell covariance matrix of its forecast vector, and $p(c)$ for the cell's transport law. Then*

$$\Delta P_c = \underbrace{\left[\|p(c) - \bar{q}_0(c)\|^2 - \|p(c) - \bar{q}_1(c)\|^2\right]}_{\text{improvement in mean-forecast fit}} - \underbrace{\left[\operatorname{tr}\mathrm{Var}(q_1 \mid c) - \operatorname{tr}\mathrm{Var}(q_0 \mid c)\right]}_{\text{increase in within-cell dispersion}}. \tag{8}$$

Only the first bracket is prior fit. The second rewards a model for making *less* dispersed predictions inside a cell at a fixed mean forecast, so ΔP can absorb an entire observed gain with both models' mean forecasts sitting exactly on the cell frequencies; Appendix M gives a four-point example. This is why the main paper calls ΔP group-level rather than prior fit, and why confidence matching moves $\widehat{\Delta P}$ so sharply: a temperature change is almost purely a change in dispersion.

B.2. Every Bregman score

Theorem 2 (Bregman-score covariance identity)**.** *Let $H_x(y) = S_\psi(q_1(\cdot \mid x), y) - S_\psi(q_0(\cdot \mid x), y)$ for a Bregman score generated by a strictly convex, differentiable ψ, and write $\Delta g_y(x) = \partial_y\psi(q_1(\cdot \mid x)) - \partial_y\psi(q_0(\cdot \mid x))$ for the contrast of the y-th gradient coordinate. Then*

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}\big(\Delta g_y(X),\, \mathbb{1}\{Y = y\} \mid C = c\big). \tag{9}$$

Corollary 1 (Quadratic and log score)**.** *For $\psi(q) = \|q\|_2^2$, $\Delta g_y(x) = 2\Delta q_y(x)$ and Eq. (9) is Eq. (3). For $\psi(q) = \sum_k q_k \log q_k$, $\Delta g_y(x) = \log q_1(y \mid x) - \log q_0(y \mid x)$, and*

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}\left(\log \frac{q_1(y \mid X)}{q_0(y \mid X)},\, \mathbb{1}\{Y = y\} \,\middle|\, C = c\right). \tag{10}$$

The log-score rows of Tab. 1 therefore carry the same exact covariance meaning as the quadratic ones. The implemented log score floors each probability at $\epsilon = 10^{-6}$; the identity is exact for the unfloored score and negligibly different whenever no relevant probability is within ϵ of zero.

B.3. One model at a time

Proposition 2 (The case-level part as a difference of within-cell reliances)**.** *For a model m, a score S and the identified cells of ϕ, let*

$$\hat{\rho}(q_m) = \frac{1}{N_*} \sum_c \sum_{i \in c} \Big[S\big(q_m(\cdot \mid x_i), y_i\big) - \frac{1}{n_c - 1} \sum_{j \in c,\, j \neq i} S\big(q_m(\cdot \mid x_i), y_j\big) \Big], \tag{11}$$

the loss in model m's score when the labels are permuted within cells. Then for the contrast $H_i(y) = S(q_1(\cdot \mid x_i), y) - S(q_0(\cdot \mid x_i), y)$, $\widehat{\Delta R} = \hat{\rho}(q_1) - \hat{\rho}(q_0)$, and the same holds for the population quantities and for any contrast that is a difference of two per-model, per-label contributions, mean recall included.

Proof. $\widehat{\Delta R} = \widehat{\Delta T} - \widehat{\Delta P}$ with $\widehat{\Delta T} = N_*^{-1} \sum_i H_i(y_i)$ and $\widehat{\Delta P} = N_*^{-1} \sum_c \sum_{i \in c} (n_c - 1)^{-1} \sum_{j \neq i} H_i(y_j)$, both linear in H; substituting $H = S(q_1) - S(q_0)$ gives the claim. □

So $\hat{\rho}$ is a within-group permutation importance [38] or model reliance [14] of whatever in the input is not the grouping variable, and a leaderboard can report it once per model. What a single model's $\hat{\rho}$ does not give is the split of a gain into its two parts, the group-level part in particular, and an interval for the difference that accounts for the two models being scored on the same relations; that is what the paired estimator and its clustered variance supply.

B.4. Coarsening, and a variable left out of the grouping

Proposition 3 (Coarsening decomposition)**.** *Let $\bar{\phi} = g(\phi)$ be any coarsening of the grouping variable, so that each cell $\bar{c}$ of $\bar{\phi}$ is a union of cells of ϕ. Then*

$$\Delta R_{\bar{c}} = \mathbb{E}\big[\Delta R_C \mid \bar{\phi} = \bar{c}\big] + \sum_{y=1}^{K} \mathrm{Cov}\big(\mathbb{E}[H_X(y) \mid C],\, \Pr(Y = y \mid C) \,\big|\, \bar{\phi} = \bar{c}\big)\,. \tag{12}$$

Proposition 4 (Sensitivity bound for an unrecorded confounder)**.** *Suppose $H_x(y) \in [-M, M]$ for all x, y (the split already requires a bounded score). Fix a cell c of ϕ, write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$ (both conditional on $\phi = c$) for the two K-vectors on the right of Eq. (12), and let*

$$\rho = \frac{\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)}{\sqrt{\sum_{y=1}^{K} \mathrm{Var}(a_y \mid c)}\sqrt{\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c)}} \in [-1, 1] \tag{13}$$

be their aggregate correlation. Then

$$\begin{aligned} \big|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\big| \;&\le\; |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(a_y \mid c)}\, \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c)} \\ &\le\; |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(H_X(y) \mid \phi = c)}\, \sqrt{\textstyle\sum_{y=1}^{K} p_y(c)\big(1 - p_y(c)\big)}, \end{aligned} \tag{14}$$

where $p_y(c) = \Pr(Y = y \mid \phi = c)$; the second inequality holds termwise ($\mathrm{Var}(a_y \mid c) \le \mathrm{Var}(H_X(y) \mid c)$ and $\mathrm{Var}(b_y \mid c) \le p_y(c)(1 - p_y(c))$ by the law of total variance, since conditioning on more variables cannot increase the variance of a conditional expectation) followed by monotonicity of the square root applied to each sum separately – a product of two square-root sums, not a sum of per-label square roots, which the law of total variance alone would not license.

Corollary 2 (Free bound)**.** *Each a_y ranges over $[-M, M]$ as a conditional expectation of H, so $\mathrm{Var}(a_y \mid c) \le M^2$ and $\sum_y \mathrm{Var}(a_y \mid c) \le KM^2$. The label probabilities satisfy $\sum_y b_y = 1$, which gives $\sum_y b_y^2 \le 1$ and, by Cauchy–Schwarz, $\sum_y (\mathbb{E}b_y)^2 \ge 1/K$; hence*

$$\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c) = \mathbb{E}\Big[\textstyle\sum_y b_y^2\Big] - \textstyle\sum_y \big(\mathbb{E}b_y\big)^2 \;\le\; 1 - \tfrac{1}{K}. \tag{15}$$

Substituting both into Eq. (14) gives the data-free bound

$$\big|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\big| \;\le\; |\rho|\, M\sqrt{K - 1}. \tag{16}$$

B.5. The estimator

Theorem 3 (Finite-sample identity)**.** *For every cell with $n_c \geq 2$, without a distributional assumption,*

$$\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c. \tag{17}$$

Moreover, when the cell's examples are drawn i.i.d. from the cell population, $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic for ΔR_c. Independence, not exchangeability alone, is what unbiasedness requires: the independent-coupling definition of ΔP_c pairs an example with a label drawn independently from the same cell, so $\mathbb{E}[H_1(Y_2)]$ matches it only for independent within-cell draws.

Proposition 5 (Exact attenuation of the in-sample plug-in)**.** *For every cell with $n_c \geq 2$ and every $i \in c$, writing $\hat{P}_i$ for Eq. (7) and $\hat{R}_i = H_i(y_i) - \hat{P}_i$,*

$$\tilde{P}_i = \hat{P}_i + \frac{1}{n_c}\hat{R}_i, \qquad \tilde{R}_i := H_i(y_i) - \tilde{P}_i = \frac{n_c - 1}{n_c}\,\hat{R}_i. \tag{18}$$

Consequently the in-sample plug-in's dataset-level statistic obeys $\widetilde{\Delta R} = N_*^{-1}\sum_{c:n_c\geq 2}(n_c-1)\widehat{\Delta R}_c$, a multiplicative shrinkage of every cell's leave-one-out covariance gain by the factor $(n_c-1)/n_c$.

B.6. Inference

With $\bar{A}_i = (n_c - 1)^{-1}\sum_{j\neq i} A_{ij}$, the Hájek influence contribution of case i is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}, \tag{19}$$

summed within each image for the one-way cluster-robust sandwich variance. A randomization test that cyclically shifts labels within every cell is a secondary, relation-level diagnostic (Appendix G).

Theorem 4 (Asymptotic normality of the covariance-gain estimator)**.** *Suppose (i) $H_x(y) \in [-M, M]$ for all x, y; (ii) examples are partitioned into clusters $g = 1, \ldots, G$ (e.g. images) that are mutually independent, with cluster sizes having a finite third moment and $\max_g |g| = o(\sqrt{G})$; (iii) the limiting variance $\sigma^2 = \lim_{G\to\infty}\mathrm{Var}\big(\sqrt{N_*}(\widehat{\Delta R} - \theta)\big)$ exists and is strictly positive, where θ is the identified-subpopulation estimand of Section F; and (iv) ϕ takes values in a fixed, finite set, each with strictly positive limiting population mass, so that every eligible cell's size n_c diverges almost surely as $N \to \infty$. Then*

$$\sqrt{N_*}\big(\widehat{\Delta R} - \theta\big) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \tag{20}$$

and the plug-in sandwich variance $\widehat{\sigma}^2 = N_*\dfrac{G}{G-1}\sum_g\big(\sum_{i\in g}\widehat{\psi}_i/N_*\big)^2$ of Section F is consistent for σ^2.

Corollary 3 (Relation to Diebold–Mariano/Giacomini–White)**.** *Under the trivial partition $\phi \equiv$ const (one cell containing every example), $\widehat{\Delta T} = N^{-1}\sum_i H_i(y_i)$ is the sample mean of the paired loss differential between the two models, and Theorem 4 applied to $\widehat{\Delta T}$ with its image-clustered interval is the grouped, cluster-robust form of the Diebold–Mariano statistic for that differential [12]; because the conditioning information set is empty, it is equivalently a degenerate instance of the Giacomini–White conditional predictive-ability test [16].*

C. A worked example

Four data points are enough to show the whole construction, and to exhibit in arithmetic the main properties Section 3 proves in general. Take one cell containing four test cases and two possible labels. Three of the cases carry label 1 and one carries label 2, so the cell's label frequencies are $(3/4, 1/4)$. The old model q_0 is uninformative within the cell: it predicts $(0.5, 0.5)$ for all four cases. We compare two candidate replacements, shown in Table 3.

Score both models with the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, and for each case tabulate the score *difference* at each of the two possible labels, not only at the observed one. Writing $h_i(y) = S(q_1(\cdot \mid x_i), y) - S(q_0(\cdot \mid x_i), y)$, all the arithmetic is in eighths:

$$h_i(y) = \begin{cases} +3/8 & \text{if } y \text{ is the label the new model moved towards,} \\ -5/8 & \text{otherwise.} \end{cases} \tag{21}$$

This vector, evaluable at labels that were never observed for case i, is the only object the method needs beyond the labels themselves.

Table 3. A four-point example in a single cell. Model A moves every case to the cell's label frequencies; model B moves each case towards its own correct label by the same amount. The two models are equally far from q_0, and only B uses information about the individual case.

case i	label y_i	$q_0(\cdot \mid x_i)$	model A: $q_1(\cdot \mid x_i)$	model B: $q_1(\cdot \mid x_i)$
1	1	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.75, 0.25)$
2	1	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.75, 0.25)$
3	1	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.75, 0.25)$
4	2	$(0.5, 0.5)$	$(0.75, 0.25)$	$(0.25, 0.75)$

Model A: a pure prior refit. Every case gets the same prediction, so $h_i(1) = +3/8$ and $h_i(2) = -5/8$ throughout, and the observed mean gain is $\Delta T = \frac{1}{4}[3 \cdot \frac{3}{8} - \frac{5}{8}] = \frac{1}{8}$. Now transport: replace each case's label in turn by each of the other three and average. Cases 1–3 are each scored against $\{1, 1, 2\}$, giving $\frac{1}{3}(\frac{3}{8} + \frac{3}{8} - \frac{5}{8}) = \frac{1}{24}$; case 4 is scored against $\{1, 1, 1\}$, giving $+\frac{3}{8}$. So $\Delta P = \frac{1}{4}[3 \cdot \frac{1}{24} + \frac{3}{8}] = \frac{1}{8}$ and $\Delta R = 0$ — exactly, in this sample, not on average over hypothetical resamples. That is the general fact: a prediction change constant within a cell contributes nothing to ΔR (Theorem 1). Note also that $\Delta P = 1/8$ is the largest transported gain any within-cell-constant prediction can earn here, since a proper score is maximised in expectation at the true frequencies $(3/4, 1/4)$ — which is what A predicts.

Model B: a case-specific improvement. Cases 1–3 are unchanged, but case 4 now moves towards label 2, so $h_4(1) = -5/8$ and $h_4(2) = +3/8$. Every case is scored at $+3/8$ on its own label, so $\Delta T = 3/8$. Under transport cases 1–3 are unaffected, but case 4 is scored against three label-1 cases at $-5/8$ apiece, so $\Delta P = \frac{1}{4}[3 \cdot \frac{1}{24} - \frac{5}{8}] = -\frac{1}{8}$ and $\Delta R = \frac{1}{2}$.

Two things follow. First, B's aggregate gain of $3/8$ *understates* its case-level improvement of $1/2$, and not because B fits the cell frequencies worse — it fits them better, moving the cell-mean prediction from 0.5 to 0.625 against a truth of 0.75, an improvement of exactly 0.09375. What offsets it is dispersion: B's forecasts vary inside the cell where the old model's did not, and by Proposition 1 the transported channel charges that variation at the same 0.09375, so the two cancel and the in-sample transported gain is zero. The $-1/8$ the estimator reports is what remains once each case is scored against the *other* cases' labels only, removing the $\frac{1}{n_c}\widehat{\Delta R} = 1/8$ the in-sample plug-in had credited to the transported channel. Second, an evaluator reading only the total would credit B with three times A's improvement, whereas the two improve on disjoint grounds: A earns $1/8$ purely by moving to the cell frequencies and gains no covariance, while B's $1/2$ of covariance is partly concealed by a transported term that turns negative. This is the failure mode that motivates the paper, in miniature — aggregate scores are not merely imprecise about composition, they can order two models by the wrong criterion.

The covariance form, and why leaving one out matters. Theorem 1 says ΔR is a within-cell covariance between the probability change and the label indicator. Computing it directly for model B gives $2\sum_y \mathrm{Cov}(\Delta q_y, \mathbb{1}\{Y = y\}) = 3/8$, not the $1/2$ transport returned. Neither is wrong: the covariance uses the cell's *own* four labels to form its reference, so each case contributes to the reference it is compared against, and that in-sample plug-in is attenuated by exactly $(n_c - 1)/n_c = 3/4$ — indeed $\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$. Transport, which scores each case only against the *other* cases' labels, removes the attenuation with no held-out data. Proposition 5 shows this holds for every cell with $n_c \geq 2$ and every grouping variable, which is why the method needs no sample splitting. The example is reproduced as a unit test in the accompanying software.

One property the example cannot show, because both candidates happen to be equally confident, is the estimator's sensitivity to confidence itself. Multiply every prediction's odds by a constant and ΔR moves, though no case has been reordered and no information added; Section 4 measures that at -0.488 in a design where the two models carry identical case-level information. Every result in this paper is therefore reported for confidence-matched pairs.

A decision the split changes. At the benchmark's own label frequencies both candidates improve and B improves three times as much, so an evaluator reading the score picks B, rightly. Suppose deployment sees label frequencies in this cell of $(\frac{1}{4}, \frac{3}{4})$ instead of the observed $(\frac{3}{4}, \frac{1}{4})$ — a label shift, the situation post-hoc prior correction addresses [26, 33]. Reweighting each case by $p'(y_i)/p(y_i)$ and rescoring sends A from $+\frac{1}{8}$ to $-\frac{3}{8}$ while leaving B at $+\frac{3}{8}$: A becomes *worse* than the model it was meant to replace, on a change that leaves every prediction exactly as it was. The decomposition says in advance which is exposed and the aggregate score does not, because A's whole gain sits in the transported channel — the one a change in label frequencies acts on directly — and B's in the covariance channel.

The covariance channel is not immune to a shift, only less exposed, and the reason is exact. Write $m_y(z) = \mathbb{E}[H_X(z) \mid Y = y, C = c]$, the expected gain at label z among cases whose true label is y. Expanding $\Delta R_c = \Delta T_c - \Delta P_c$ over label

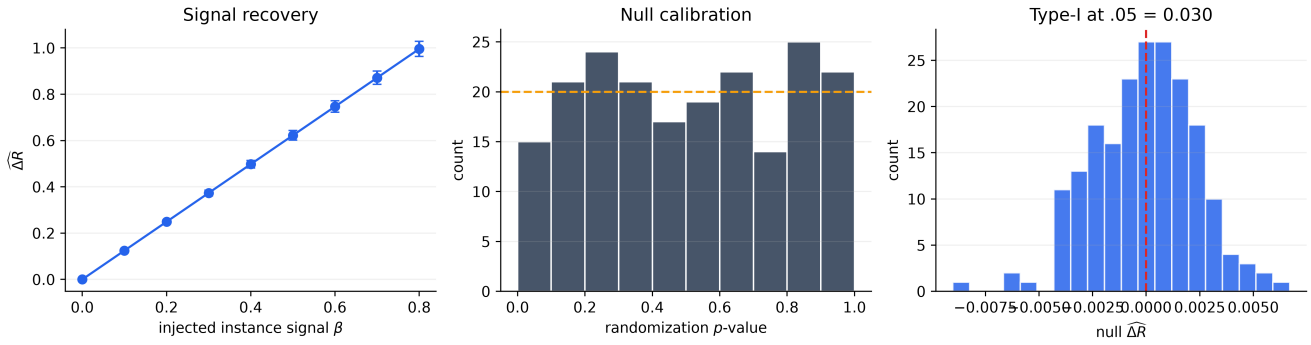


Figure 4. **Controlled validation.** The split recovers injected within-cell signal monotonically (mean ± one standard deviation), produces approximately uniform null randomization p-values, and controls type-I error in 200 null runs.

pairs gives

$$\Delta R_c = \sum_{y<z} p(y)\, p(z)\, D(y,z), \qquad D(y,z) = m_y(y) - m_y(z) + m_z(z) - m_z(y). \tag{22}$$

A label shift changes $p(y)$ and leaves $p(x \mid y)$, hence every $m_y(z)$ and every D, fixed. With two labels the sum has one term, so a shift rescales ΔR_c by $p(1)p(2)$ and cannot change its sign; in the example above $p(1)p(2) = \frac{3}{16}$ both before and after, which is why B's covariance gain is untouched. With three or more labels the pairwise contrasts can differ in sign, and then a shift can reverse the covariance gain. Concretely: three case types, each always carrying its own label, the old model uniform, and the new model predicting $(0.65, 0.35, 0)$, $(0.65, 0.05, 0.30)$ and $(0.20, 0.70, 0.10)$ for the three types gives $D = -0.6$, $+1.1$ and -1.7. With the labels balanced, 40 cases each, the estimated covariance gain is -0.13445; shifting the counts to $(40, 2, 78)$, which starves the two negative contrasts of weight, makes it $+0.21840$. `tests/test_antisymmetric.py` asserts both values, and asserts that no two-label cell reverses sign. The operational content of the split is therefore this: $\widehat{\Delta P}$ is the part of a reported improvement that a change in label frequencies acts on directly, and $\widehat{\Delta R}$ is a frequency-weighted sum of fixed pairwise contrasts, which a shift re-weights but, in a two-label cell, cannot reverse. Section 6 exhibits the same structure on real models.

D. Coarsening and unrecorded covariates

Appendix N.4 reports that replacing the class-pair cell with a coarser subject-only cell changes the estimated covariance gain. The following result gives that sensitivity an exact population-level mechanism rather than treating it as an empirical curiosity.

The first term is the coarse cell's share of the fine-grained covariance gain; the second is a between-cell covariance, over the finer cells nested inside $\bar{c}$, between each fine cell's typical gain at label y and its label-y frequency. It is not sign definite, so coarsening ϕ can inflate or deflate the credited covariance gain, and it vanishes exactly when either quantity is constant within each coarse cell. This is why the finest defensible ϕ is the conservative default: coarsening adds cross-cell prior structure into ΔR, while the finer partition's gain is preserved on average through the first term. The proof is a per-label application of the law of total covariance (Appendix E).

D.1. Sensitivity to an unrecorded confounder

Read in one direction, Proposition 3 says what folding a covariate into ϕ does. Read in the other, it quantifies what happens when a relevant covariate is *never* declared, which is the estimator's main scope limit. Let Z be unrecorded and let (ϕ, Z) be the finer, fully adjusted cell. Applying Proposition 3 with $\bar{\phi} = \phi$ gives, for every declared cell c,

$$\Delta R_c(\phi) = \mathbb{E}\big[\Delta R_{(\phi,Z)} \mid \phi = c\big] + \sum_{y=1}^{K} \mathrm{Cov}\big(\mathbb{E}[H_X(y) \mid \phi, Z],\, \Pr(Y = y \mid \phi, Z) \mid \phi = c\big)\,, \tag{23}$$

whose first term is what an analyst who could observe Z would report and whose second is the bias of using ϕ alone. The identity is exact but needs Z's joint law with (H, Y). The following bound replaces that with a judgement about how strongly an unrecorded variable could plausibly act, in the spirit of Rosenbaum [32].

Proposition 4 states that bound and Corollary 2 its data-free form; both are proved in Appendix E. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds ratio, it can be elicited the same way across benchmarks. Appendix I evaluates both forms on our own estimates, and the sharp one is the less comfortable: an unrecorded covariate correlated at 0.061 would account for the entire covariance gain reported in Appendix N.2.

E. Proofs

E.1. Proof of Theorem 1

Condition throughout on $C = c$. Expanding over labels gives

$$\Delta T_c = \sum_y \mathbb{E}[H_X(y)\mathbb{1}\{Y = y\}], \qquad \Delta P_c = \sum_y \mathbb{E}[H_X(y)] \Pr(Y = y).$$

Subtracting yields the covariance sum in Eq. (2); the first identity is the definition of ΔR_c. Under the quadratic score,

$$H_X(y) = 2\Delta q_y(X) - \{\|q_1(\cdot \mid X)\|_2^2 - \|q_0(\cdot \mid X)\|_2^2\}.$$

The bracketed term is independent of the label index. Its contribution summed over y is $\mathrm{Cov}(B(X), \sum_y \mathbb{1}\{Y = y\}) = \mathrm{Cov}(B(X), 1) = 0$. The remaining terms give Eq. (3). If $H_X(y) = H_c(y)$ almost surely, every covariance is zero. □

E.2. Proof of Proposition 1

Condition on $C = c$ and let $Y' \sim p(c)$ be independent of X. For the quadratic score $S_{\mathrm{B}}(q, y) = 2q_y - \|q\|^2$,

$$\mathbb{E}\big[S_{\mathrm{B}}(q_m, Y') \mid c\big] = 2\big\langle p(c), \bar{q}_m(c)\big\rangle - \mathbb{E}\big[\|q_m\|^2 \mid c\big]. \tag{24}$$

Decompose the second moment as $\mathbb{E}[\|q_m\|^2 \mid c] = \|\bar{q}_m(c)\|^2 + \mathrm{tr}\,\mathrm{Var}(q_m \mid c)$, and complete the square using $2\langle p, \bar{q}\rangle - \|\bar{q}\|^2 = \|p\|^2 - \|p - \bar{q}\|^2$:

$$\mathbb{E}\big[S_{\mathrm{B}}(q_m, Y') \mid c\big] = \|p(c)\|^2 - \|p(c) - \bar{q}_m(c)\|^2 - \mathrm{tr}\,\mathrm{Var}(q_m \mid c). \tag{25}$$

Subtracting the $m = 0$ expression from the $m = 1$ expression, the $\|p(c)\|^2$ terms cancel and Eq. (8) follows, since $\Delta P_c = \mathbb{E}[S_{\mathrm{B}}(q_1, Y') \mid c] - \mathbb{E}[S_{\mathrm{B}}(q_0, Y') \mid c]$ by definition. □

E.3. Proof of Theorem 2

Condition on $C = c$. The proof of Theorem 1 above establishes $\Delta R_c = \sum_y \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)$ for *any* score contrast H, not only the quadratic one; only the identification of $H_x(y)$ with a specific functional form is score-dependent. For the Bregman score of the Bregman score $S_\psi(q, y) = \psi(q) + \langle \nabla\psi(q), e_y - q\rangle$,

$$H_x(y) = \underbrace{\big[\psi(q_1(x)) - \psi(q_0(x))\big] - \big[\langle \nabla\psi(q_1(x)), q_1(x)\rangle - \langle \nabla\psi(q_0(x)), q_0(x)\rangle\big]}_{=:B(x)} + \Delta g_y(x),$$

writing $q_i(x)$ for $q_i(\cdot \mid x)$, where $B(x)$ collects every term not indexed by y and $\Delta g_y(x) = \partial_y\psi(q_1(x)) - \partial_y\psi(q_0(x))$ is the y-indexed remainder, exactly as in the proof of Theorem 1 for the quadratic case. Substituting,

$$\sum_y \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid c) = \mathrm{Cov}\Big(B(X), \sum_y \mathbb{1}\{Y = y\} \,\Big|\, c\Big) + \sum_y \mathrm{Cov}(\Delta g_y(X), \mathbb{1}\{Y = y\} \mid c).$$

The first term is $\mathrm{Cov}(B(X), 1 \mid c) = 0$ since $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The second term is Eq. (9). □

E.4. Proof of Theorem 3

Summing the kernel in Eq. (6) over unordered pairs, every observed term $H_i(y_i)$ occurs in $n_c - 1$ pairs with coefficient $1/2$, while the two crossed terms together enumerate every ordered pair $H_i(y_j)$, $i \neq j$, with coefficient $1/2$. Since $\binom{n_c}{2} = n_c(n_c - 1)/2$,

$$\widehat{\Delta R}_c = \frac{1}{n_c}\sum_i H_i(y_i) - \frac{1}{n_c(n_c - 1)}\sum_{i \neq j} H_i(y_j) = \widehat{\Delta T}_c - \widehat{\Delta P}_c.$$

This is deterministic. When the cell's examples are i.i.d. draws from the cell population, the average of the symmetric order-two kernel is the standard unbiased U-statistic for its population expectation [18]; independence is used, not merely exchangeability, because ΔP_c is defined through an independent label coupling, so $\mathbb{E}[H_1(Y_2)] = \mathbb{E}[H_X(Y')]$ requires the two draws to be independent. □

E.5. Proof of Proposition 3

Fix y and condition throughout on $\bar{\phi} = \bar{c}$. The law of total covariance applied to the pair $(H_X(y), \mathbb{1}\{Y = y\})$ with the finer conditioning variable C nested inside $\bar{c}$ gives

$$\begin{aligned}\operatorname{Cov}\big(H_X(y), \mathbb{1}\{Y=y\} \mid \bar{\phi} = \bar{c}\big) = \mathbb{E}\big[\operatorname{Cov}(H_X(y), \mathbb{1}\{Y=y\} \mid C) \,\big|\, \bar{\phi} = \bar{c}\big] \\ + \operatorname{Cov}\big(\mathbb{E}[H_X(y) \mid C],\, \Pr(Y = y \mid C) \,\big|\, \bar{\phi} = \bar{c}\big)\,,\end{aligned}$$

using $\mathbb{E}[\mathbb{1}\{Y = y\} \mid C] = \Pr(Y = y \mid C)$. Summing over $y = 1, \ldots, K$ and applying the population identity of Theorem 1 to the left side (at $\bar{\phi} = \bar{c}$) and to the first term on the right side (at $C = c$, then averaged over c within $\bar{c}$) yields Eq. (12). □

E.6. Proof of Proposition 5

Fix a cell c with $n_c \geq 2$ and $i \in c$. By definition of the label count, $\sum_{y=1}^{K} m_{cy} H_i(y) = \sum_{j \in c} H_i(y_j) = H_i(y_i) + \sum_{j \neq i} H_i(y_j)$, and Eq. (7) gives $\sum_{j \neq i} H_i(y_j) = (n_c - 1)\hat{P}_i$. Hence

$$\tilde{P}_i = \frac{1}{n_c} \sum_{y} m_{cy} H_i(y) = \frac{H_i(y_i) + (n_c - 1)\hat{P}_i}{n_c}.$$

Substituting $H_i(y_i) = \hat{P}_i + \hat{R}_i$ gives $\tilde{P}_i = \hat{P}_i + n_c^{-1}\hat{R}_i$, and $\tilde{R}_i = H_i(y_i) - \tilde{P}_i = \hat{R}_i - n_c^{-1}\hat{R}_i = (n_c - 1)n_c^{-1}\hat{R}_i$. Averaging $\tilde{R}_i$ within cell c and reweighting by n_c across cells, as in the dataset-level definition of $\widehat{\Delta Z}$, gives $\widehat{\Delta R} = N_*^{-1} \sum_c n_c \cdot \frac{n_c - 1}{n_c} \widehat{\Delta R}_c = N_*^{-1} \sum_c (n_c - 1) \widehat{\Delta R}_c$. □

E.7. Proof of Proposition 4

Fix $\phi = c$ throughout and write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$, both random through their dependence on (ϕ, Z). Let $U = (a_y - \mathbb{E}[a_y \mid c])_{y=1}^{K}$ and $V = (b_y - \mathbb{E}[b_y \mid c])_{y=1}^{K}$ be the corresponding mean-zero $\mathbb{R}^K$-valued random vectors conditional on $\phi = c$. The map $(U, V) \mapsto \mathbb{E}[\langle U, V \rangle \mid c] = \sum_y \operatorname{Cov}(a_y, b_y \mid c)$ is an inner product on the space of mean-square-integrable $\mathbb{R}^K$-valued random variables conditional on $\phi = c$, so by the Cauchy–Schwarz inequality applied to that inner product,

$$\Big|\sum_{y=1}^{K} \operatorname{Cov}(a_y, b_y \mid c)\Big| \leq \sqrt{\mathbb{E}[\|U\|^2 \mid c]}\,\sqrt{\mathbb{E}[\|V\|^2 \mid c]} = \sqrt{\textstyle\sum_y \operatorname{Var}(a_y \mid c)}\,\sqrt{\textstyle\sum_y \operatorname{Var}(b_y \mid c)}.$$

By the definition of ρ, the left side equals $|\rho|\sqrt{\sum_y \operatorname{Var}(a_y \mid c)}\sqrt{\sum_y \operatorname{Var}(b_y \mid c)}$ exactly, so this step alone is not yet a bound; two further steps supply one, at two different levels of conservatism. First, $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$ is itself a conditional expectation of $H_X(y)$ given more information than $\phi = c$ alone, so the law of total variance gives $\operatorname{Var}(H_X(y) \mid c) = \mathbb{E}[\operatorname{Var}(H_X(y) \mid \phi, Z) \mid c] + \operatorname{Var}(a_y \mid c) \geq \operatorname{Var}(a_y \mid c)$; likewise $b_y = \Pr(Y = y \mid \phi, Z)$ satisfies $\operatorname{Var}(b_y \mid c) \leq \operatorname{Var}(\mathbb{1}\{Y = y\} \mid c) = p_y(c)(1 - p_y(c))$ by the same argument applied to $\mathbb{1}\{Y = y\}$ in place of $H_X(y)$. Both bounds hold termwise, so summing over y and applying the (monotone) square root to each sum separately gives the observable, per-cell bound

$$\sqrt{\textstyle\sum_y \operatorname{Var}(a_y \mid c)}\sqrt{\textstyle\sum_y \operatorname{Var}(b_y \mid c)} \leq \sqrt{\textstyle\sum_y \operatorname{Var}(H_X(y) \mid c)}\sqrt{\textstyle\sum_y p_y(c)(1 - p_y(c))},$$

the second inequality of Eq. (14); this is a bound on a product of two sums, and does not simplify to a sum of per-label square roots, since Cauchy–Schwarz runs the other way for that rearrangement. Second, for Corollary 2, $a_y \in [-M, M]$ as a conditional expectation of a $[-M, M]$-valued variable, so Popoviciu's inequality gives $\operatorname{Var}(a_y \mid c) \leq M^2$ for every y and hence $\sum_y \operatorname{Var}(a_y \mid c) \leq KM^2$. For the b_y the same argument would give $\operatorname{Var}(b_y \mid c) \leq 1/4$ and a total of $K/4$, but that treats the b_y as K unconstrained quantities in $[0, 1]$ when in fact $\sum_y b_y = 1$ almost surely. Using that constraint,

$\sum_y b_y^2 \le \big(\sum_y b_y\big)^2 = 1$ pointwise, while $\sum_y (\mathbb{E}b_y)^2 \ge K^{-1}\big(\sum_y \mathbb{E}b_y\big)^2 = 1/K$ by Cauchy–Schwarz, so

$$\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c) = \mathbb{E}\Big[\sum_y b_y^2\Big] - \sum_y \big(\mathbb{E}b_y\big)^2 \le 1 - \frac{1}{K},$$

which is Eq. (15) and is strictly smaller than $K/4$ for every $K > 2$. Therefore

$$\Big|\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)\Big| \le |\rho|\sqrt{KM^2}\sqrt{1 - 1/K} = |\rho|\, M\sqrt{K-1}.$$

By Eq. (12), the left side of each display is exactly $|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]|$, giving Eq. (14) and Eq. (16). □

F. Influence function: estimand and derivation

The estimand under singleton exclusion. Cells with $n_c = 1$ identify no within-cell counterfactual and are excluded, so the dataset-level statistic targets the *identified-subpopulation* quantity: writing π_c for the cell probabilities, the target of $\widehat{\Delta R}$ is the π-weighted mean $\theta = \sum_c \pi_c \Delta R_c / \sum_c \pi_c$ taken over cells that appear with at least two examples. At finite N the identified set is random; we treat the interval as conditional on identification and report the identified fraction alongside every estimate (99.0% of relations on VG150, 100% on CIFAR-100-LT), so the gap between the conditional and full-population targets is bounded by the reported coverage.

Derivation of Eq. (19). Fix a cell c and let $\theta_c = \Delta R_c$ with $g_c(z) = \mathbb{E}[A(z, Z') \mid Z' \in c]$ the kernel projection. The classical Hájek projection of the order-two U-statistic $\widehat{\Delta R}_c$ is

$$\widehat{\Delta R}_c = \theta_c + \frac{2}{n_c}\sum_{i \in c}\big(g_c(Z_i) - \theta_c\big) + O_p(n_c^{-1}).$$

The dataset statistic weights cells by their example counts, $\widehat{\Delta R} = N_*^{-1}\sum_c n_c \widehat{\Delta R}_c$, and cell membership is itself random (multinomial over cells), so an example i contributes both through the within-cell projection and through the between-cell composition term $\theta_{C_i} - \theta$. Combining the two sources,

$$\widehat{\Delta R} - \theta = \frac{1}{N_*}\sum_i \Big[2\big(g_{C_i}(Z_i) - \theta_{C_i}\big) + \big(\theta_{C_i} - \theta\big)\Big] + o_p(N_*^{-1/2}),$$

where the dataset-level remainder requires one condition beyond the per-cell expansion, because C per-cell remainders do not aggregate for free. Writing $C = \#\{c : n_c \ge 2\}$ for the number of eligible cells, the weighted sum of the per-cell degenerate terms is $N_*^{-1}\sum_c n_c \cdot O_p(n_c^{-1}) = O_p(C/N_*)$, which is $o_p(N_*^{-1/2})$ provided

$$C = o\big(\sqrt{N_*}\big). \tag{26}$$

This is what makes the second-order rate $O_p(n_c^{-1})$ rather than $o_p(n_c^{-1/2})$ worth stating: the sharper per-cell rate is what allows a condition on the cell *count* alone, with no further requirement on individual cell sizes. On VG150, $C = 6{,}346$ against $N_* = 227{,}337$, so $C/\sqrt{N_*} \approx 13.3$ — the condition is a statement about the limit, and at these finite sample sizes it is the coverage simulation of Section 4, not Eq. (26), that supports the reported intervals. Estimating $g_{C_i}(Z_i)$ by the pair-kernel mean $\bar{A}_i$, θ_c by $\widehat{\Delta R}_c$, and θ by $\widehat{\Delta R}$ gives the plug-in influence contribution $\widehat{\psi}_i = 2(\bar{A}_i - \widehat{\Delta R}_c) + (\widehat{\Delta R}_c - \widehat{\Delta R}) = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}$, which is Eq. (19) and has exactly zero sample mean by construction. Its empirical 95% coverage in a VG-like regime (image-clustered examples, heavy-tailed cell sizes with a large mass of near-minimum cells) is validated in Section 4.

Proof of Theorem 4. The linearization above gives $\widehat{\Delta R} - \theta = N_*^{-1}\sum_i \psi_i + o_p(N_*^{-1/2})$, under Eq. (26), for the population influence contribution $\psi_i = 2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta)$, which has zero mean by construction and satisfies $|\psi_i| \le C(M)$ for an absolute constant $C(M)$ depending only on the score bound M in condition (i), since g_{C_i}, θ_{C_i}, and θ are themselves

averages of the bounded kernel $A_{ij} \in [-2M, 2M]$. Because clusters are mutually independent (condition (ii)), $T_g = \sum_{i \in g} \psi_i$ for $g = 1, \ldots, G$ are independent, mean-zero random variables with $|T_g| \le |g|\, C(M)$. Condition (ii)'s finite third moment for cluster sizes and $\max_g |g| = o(\sqrt{G})$ give $\sum_g E|T_g|^3 \le C(M)^3 \sum_g \mathbb{E}|g|^3 = O(G)$ while $\left(\sum_g \mathrm{Var}(T_g)\right)^{3/2} = \Theta(G^{3/2})$ under condition (iv), so Lyapunov's ratio $\sum_g E|T_g|^3 / (\sum_g \mathrm{Var}(T_g))^{3/2} \to 0$; the Lyapunov condition implies the Lindeberg condition, and the Lindeberg–Feller central limit theorem for triangular arrays of independent summands [35, 43] gives $\left(\sum_g T_g\right) / \sqrt{\sum_g \mathrm{Var}(T_g)} \xrightarrow{d} \mathcal{N}(0,1)$. Condition (iii) makes $N_* / \sum_g \mathrm{Var}(T_g) \to 1/\sigma^2$ in probability by definition of σ^2, and the $o_p(N_*^{-1/2})$ remainder is asymptotically negligible after multiplying by $\sqrt{N_*}$, so $\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ by Slutsky's theorem. For the variance estimator, $\widehat{\Delta R} \to_p \theta$ by the weak law of large numbers across the diverging number of identified examples, and $\widehat{\Delta R}_c \to_p \Delta R_c$ for each eligible cell by the weak law of large numbers applied within that cell – this is where condition (iv) is used, and it is not implied by anything aggregate: (iv) guarantees $n_c \to \infty$ almost surely for every cell in the fixed, finite support of ϕ, which is what a within-cell WLLN requires, whereas a condition on the total identified count bounds no individual cell's size. So $\widehat{\psi}_i \to_p \psi_i$ pointwise, and $\widehat{\sigma}^2$ – a sum of squares of cluster-aggregated $\widehat{\psi}_i$, normalized by N_* and G – is a continuous functional of these consistent plug-ins, hence $\widehat{\sigma}^2 \to_p \sigma^2$ by the continuous mapping theorem. □

Efficient computation. The implementation does not instantiate all pairs. In addition to Eq. (7), let $G_c(y) = \sum_{j \in c} H_j(y)$ and $O_c = \sum_{j \in c} H_j(y_j)$. The pair-kernel mean for observation i is

$$\bar{A}_i = \frac{1}{2}\left[H_i(y_i) + \frac{O_c - H_i(y_i)}{n_c - 1} - P_i - \frac{G_c(y_i) - H_i(y_i)}{n_c - 1}\right].$$

Therefore both the point estimate and Hájek projection require only label counts, column sums, and matrix–vector products inside each cell. For image clusters g, the implemented variance is

$$\widehat{\mathrm{Var}}(\widehat{\Delta R}) = \frac{G}{G-1}\frac{1}{N_*^2}\sum_{g=1}^{G}\left(\sum_{i \in g}\widehat{\psi}_i\right)^2,$$

where G is the number of images and $\widehat{\psi}_i$ is Eq. (19). We use a normal critical value; with 27,955 image clusters, small-G corrections are immaterial.

G. Randomization algorithm

For each cell, sort examples once and index them $0, \ldots, n_c - 1$. Randomization b draws $o_c^{(b)}$ uniformly from this index set and maps label position r to $(r + o_c^{(b)}) \bmod n_c$. Independent offsets across cells define an element of the product of cyclic groups. The cell label histogram is unchanged, so $\sum_y m_{cy} H_i(y)$ in Eq. (7) is precomputed; only the assigned label lookup changes. One randomization is $O(N)$ after the $O(NK)$ setup. The plus-one Monte Carlo correction includes the observed assignment and prevents zero p-values.

H. Implementation and reproducibility

The NumPy implementation is in `gainsplit/antisymmetric.py`. The primary routine `decompose_gain` validates probability matrices, forms score contrasts, excludes singleton cells, computes all three components, verifies the identity numerically, and returns per-instance transported/covariance values plus intervals. The Visual Genome driver is `experiments/run_sgg_gainsplit.py`; controlled validation and sensitivity checks are in `experiments/antisymmetric_simulation.py` and `experiments/antisymmetric_ablations.py`. All random seeds are fixed. Unit tests cover the decomposition identity, quadratic covariance identity, cell-constant null, singleton reporting, cluster inference, randomization power, and log-score path, plus direct numerical checks of Propositions 5 and 3 (the latter against an explicit discrete population with exact expectations, independent of the estimator's own code path), Corollary 1's log-score covariance identity, Corollary 3's equivalence to the (unclustered) Diebold–Mariano statistic at a trivial grouping variable, and Proposition 4's bound ordering (exact bias ≤ tight, per-cell bound ≤ crude, data-free bound) on a second explicit discrete population with a known omitted confounder. The covariance-identity cross-check quoted in Section N.5 is reproduced by

Algorithm 1 Gain decomposition by label transport

Require: frozen q_1, q_0; labels y; cells ϕ; proper score S
Ensure: $\widehat{\Delta T}, \widehat{\Delta P}, \widehat{\Delta R}$ and uncertainty
$H_i(k) \leftarrow S(q_{1i}, k) - S(q_{0i}, k)$ for all i, k
for each cell c with $n_c \geq 2$ **do**
  count labels $m_{c1}, \ldots, m_{cK}$
  for each $i \in c$ **do**
    $P_i \leftarrow (\sum_k m_{ck} H_i(k) - H_i(y_i))/(n_c - 1)$
    $R_i \leftarrow H_i(y_i) - P_i$
  end for
end for
$\widehat{\Delta T} \leftarrow \text{mean } H_i(y_i)$; $\widehat{\Delta P} \leftarrow \text{mean } P_i$; $\widehat{\Delta R} \leftarrow \text{mean } R_i$
compute image-clustered Hájek interval and cell-shift randomization p

`experiments/cifar_covariance_check.py`; the calibration-matched protocol and the prior-matching experiment are `experiments/cifar_recalibration_control.py` and `experiments/vg_prior_consequence.py`. Table 4's free-bound rows are reproduced by `experiments/sensitivity_bound_check.py` and its sharp rows, including the robustness value $\rho^\dagger$ of Eq. (27), by `experiments/sensitivity_analysis.py`; the evaluation of Corollary 2 is `experiments/sensitivity_bound_check.py`, which recomputes the crude bound directly from the committed `antisymmetric_ablations.json` rather than re-running the decomposition; `experiments/sensitivity_analysis.py` computes Proposition 4's tighter, per-cell form from per-instance arrays (Section H.1).

Full-data VG predictors (Sections N.1–N.4). FREQ uses add-one smoothing over the 50 predicates within each training class pair, backing off to the subject-marginal and then the global predicate histogram for pairs unseen in training. The MLPs consume fixed random class embeddings (32 dimensions per class, seed 0, concatenated for subject and object) on standardized features and train with AdamW (learning rate 10^{-3}, weight decay 10^{-5}, batch size 4096): MLP-CLASS with hidden sizes $(256, 128)$ for 12 epochs, MLP-SPATIAL with $(256, 128)$ for 15 epochs, and MLP-SPATIAL+ with $(512, 256, 128)$ for 20 epochs – the "+" variant differs in both width and schedule, as noted in Section 4.

Data availability. The repository ships the estimator, every driver script, and the cached results files that every printed number traces to (`results/*.json`). The cached per-relation probability arrays that the drivers consume (`sgg_dists.npz`, `vg_visual_models.npz`, `cifar_lt_results.npz`) are regenerable from the committed training scripts with fixed seeds and are archived alongside the code release, so exact reproduction does not require retraining.

H.1. Compute environment for Sections N.3–N.5

Model training and feature extraction for both new experiments ran on a single NVIDIA T4 (15 GB) GPU. The decomposition itself is pure NumPy and requires no accelerator: it consumes the resulting cached probability arrays, as in Section 4. All Python code runs under a pinned environment (Python 3.13).

Matched-subsample models (Section 6). The three compared predictors share a fixed, seeded (seed $= 0$) subsample of 100,000 of the 532,987 training relations, drawn without replacement uniformly at the relation level (not the image level); the full 229,605-relation test split is always used for evaluation. All three use the identical head and optimizer – AdamW, learning rate 10^{-3}, weight decay 10^{-5}, 15 epochs, batch size 4096, hidden sizes $(256, 128)$, on standardized features – so they differ only in their input features.

Subject/object crops are taken from the annotated boxes and CLIP-preprocessed (bicubic resize to 224×224 with CLIP's published normalization constants); a crop with no valid overlap with the image falls back to a blank crop rather than being dropped, so every relation contributes a feature row. CLIP runs in half precision with a batch size of 512 crops and is never fine-tuned. Two implementation details are worth recording. First, the pretrained OpenAI weights were trained with the QuickGELU activation, so the encoder must be instantiated as `ViT-B-32-quickgelu`; loading the same weights under the plain `ViT-B-32` configuration silently substitutes a standard GELU and degrades every embedding. Second, because an annotated object box participates in several relations, crops are deduplicated by (image, box) before encoding and gathered

afterwards: the 659,210 crops implied by the audited relations collapse to 422,143 unique ones, removing 36.0% of the encoder work without changing any feature. The 71,990 images those relations reference were all available in the run behind the reported predictions (an earlier run fetching them one at a time missed three, which fell back to a blank crop); the predictions are archived with the code.

ResNet-32 triple (Section N.5). CIFAR-100-LT uses the standard exponential-profile construction of Cui et al. [8] at imbalance ratio 100: class c (zero-indexed, $0, \dots, 99$) keeps $n_c = \lfloor 500 \cdot \mu^c \rfloor$ training images, where $\mu = 100^{-1/99}$, so class 0 keeps all 500 images and class 99 keeps 5; the balanced 10,000-image test split is untouched. All three models are a from-scratch, randomly initialized CIFAR ResNet-32 (three stages of five basic residual blocks, 16/32/64 channels) trained for 80 epochs with SGD (momentum 0.9, Nesterov, weight decay 2×10^{-4}), batch size 128, standard crop-and-flip augmentation, and a cosine learning-rate schedule (peak 0.1, five-epoch linear warmup). The only difference between the runs is the per-example loss weight. CE uses uniform weights throughout. CB applies the effective-number class-balanced weight $w_c \propto (1-\beta)/(1-\beta^{n_c})$ with $\beta = 0.9999$, renormalized to average to one across classes, from initialization. DRW trains with uniform weights until epoch 60 and switches to exactly the CB weights for the final twenty epochs [3]; no other hyperparameter changes at the switch. The robustness matrix of Table 12 repeats this recipe verbatim at seeds 1 and 2 (ratio 100) and at ratios 10 and 50 (seed 0), re-drawing the long-tailed subset with the configuration's own seed and initializing all three arms of a configuration identically (`experiments/colab_cifar_multiseed.py`). The post-hoc LA arm is not trained: it rescales the baseline's test probabilities by the inverse training class prior, $q_{\mathrm{LA}}(y \mid x) \propto q_{\mathrm{CE}}(y \mid x)/\pi_{\mathrm{train}}(y)$, renormalized.

I. Sensitivity bounds on the reported estimates

What the bound says about this paper's own estimates. Table 4 reports both forms of the bound on the Visual Genome pair of Section N.2, together with the coarsening from the class-pair to the subject-only cell, which instantiates Eq. (12) exactly with $Z =$ object class. The free bound is directionally correct and, at $K = 50$, still 1,887 times the actual coarsening bias, so any $\bar{\rho}$ an analyst would plausibly declare certifies a limit far above what is observed. That looseness is a property of a worst-case inequality applied to label probabilities that concentrate away from the extremes at which it binds, not a defect of this dataset — but it also means the free bound cannot support a claim about whether any particular reported $\widehat{\Delta R}$ would survive confounding.

The sharp per-cell form of Eq. (14), which replaces the worst-case sums by the cell's own $\sum_y \mathrm{Var}(H_X(y) \mid c)$ and $\sum_y p_y(c)(1-p_y(c))$, can support such a claim, and it is less reassuring than the free bound. Computed by `experiments/sensitivity_analysis.py` from per-instance arrays for the same pair, it gives $B = 0.23284$ at $\bar{\rho} = 1$, so the *robustness value* — the smallest aggregate correlation at which the bound reaches the estimate — is (its estimate, 0.01427, is computed from those arrays and differs from the 0.01439 of the tables by 0.00012, under 1%, within the interval's half-width)

$$\rho^{\dagger} = \frac{|\widehat{\Delta R}|}{B} = \frac{0.01427}{0.23284} = 0.06127. \tag{27}$$

An unrecorded covariate whose aggregate correlation with the within-cell gain and the within-cell label distribution reaches 0.061 is enough to account for the whole of that $\widehat{\Delta R}$. At the $\bar{\rho} = 0.1$ that this section suggests as a plausible elicited value, the sharp bound is 0.02328, which *exceeds* the estimate. We therefore do not claim that a confounder would have to be implausibly strong to explain these Visual Genome gains away; on this evidence a modest one would suffice, and the reader should treat Section N.2's covariance gains as conditional on the declared ϕ being close to complete. The flagship audit of Section 5 is exposed differently: its reported $\widehat{\Delta R}$ is indistinguishable from zero, but by Proposition 3 the omitted between-cell term has either sign, so a null at the declared ϕ can hide a non-zero part at a finer grouping; the bound applies to it in both directions.

J. Conditions for the limit theorem

Two things about condition (iv) deserve saying plainly rather than in a footnote. It is what licenses treating the per-cell plug-ins $\widehat{\Delta R}_c$ as consistent for ΔR_c in the variance argument, and a cell's size is not controlled by the aggregate identified count, so this needs its own divergence requirement. But it also forces every cell to be eventually identified, hence $N_*/N \to 1$, so no separate assumption on the identified fraction is needed, and the identified-subpopulation estimand θ coincides with the unconditional target. The price is that the theorem does not describe the regime the paper's own applications sit in, where

Table 4. Both forms of the Proposition 4 bound on the MLP-SPATIAL vs. MLP-CLASS pair, and the coarsening already reported in Table 10 ($K = 50$, $M = 2$ for the quadratic score). The empirical bias is the law-of-total-expectation aggregate of Eq. (12)'s covariance term across all subject-only cells. `experiments/sensitivity_bound_check.py` reproduces the free-bound rows from the committed ablation results and `experiments/sensitivity_analysis.py` the sharp rows from per-instance arrays.

Quantity	Value
$\widehat{\Delta R}$, $\phi =$ subject-only	0.02181
$\widehat{\Delta R}$, $\phi =$ class-pair	0.01439
Empirical coarsening bias	0.00742
Free bound, $M\sqrt{K-1}$, at $\bar{\rho} = 1$	14.00
Minimum $\bar{\rho}$ for the free bound to reach the bias	0.00053
Sharp bound B at $\bar{\rho} = 1$	0.23284
Sharp bound at $\bar{\rho} = 0.1$	0.02328
Robustness value $\rho^\dagger$	0.06127

θ is a random, N-dependent target because coverage is short of one (99.0% on VG150, 98.9% in Section 5). What the theorem licenses is the interval's form and its variance estimator; what licenses reading the reported intervals at finite N is the coverage simulation of Section 4, and Section 7 states the gap between the two rather than eliding it.

The proof (Section E) combines the exact linearization of Section F with a Lindeberg–Feller central limit theorem for the independent-across-cluster sum of mean-zero, uniformly bounded terms [35, 43]; boundedness (i) makes the Lindeberg condition automatic given (ii), and consistency of $\widehat{\sigma}^2$ follows from the weak law of large numbers applied to each plug-in $\widehat{\Delta R}_c$, $\widehat{\Delta R}$ together with continuity of the variance functional. One step needs more care than that section's per-cell statement supplies: a per-cell Hájek remainder of order $o_p(n_c^{-1/2})$ does not aggregate to $o_p(N_*^{-1/2})$ at the dataset level when the number of cells grows. The second-order degenerate term of each cell's U-statistic is $O_p(n_c^{-1})$, so the weighted sum of these terms is $O_p(C/N_*)$ with $C = \#\{c : n_c \geq 2\}$, and it is negligible at the $\sqrt{N_*}$ rate exactly when $C = o(\sqrt{N_*})$, Eq. (26). Under condition (iv) the set of cells is fixed, so this holds trivially; for a number of cells that grows with N it is the condition to state. VG150 does not meet it at its sample size ($C/\sqrt{N_*} \approx 13.3$), so the theorem does not by itself license the reported intervals. Nor is the omitted term small in every cell: in a cell of two cases the U-statistic is its single kernel, and the degenerate term is of the same order as the projection the variance keeps. Table 5 measures what this costs. With every cell holding two cases (each relation its own image, cells and their label priors fixed across replicates) the nominal 95% interval covers 81.5% and the standard error is 0.71 of the estimator's standard deviation; from three cases up coverage is nominal, and with VG150's own cell sizes it is 96.3%. On VG150 1.0% of audited relations sit in two-case cells; a grouping finer than the benchmark's would need a variance with the second-order term, or a delete-a-group jackknife over images (used in Appendix Q.3).

The mean-recall split scores a hit matrix, and its group-level share is a ratio whose interval comes from that jackknife; neither was covered above. A second simulation (`sim_hit_coverage.py`) draws cells with VG150's cell-size distribution (6,066 relations in 191 cells), groups relations into 867 images that cut across cells, and gives each case per-label evidence $e_x(y) = \mathbb{1}[y = y_x]\zeta_x + \varepsilon$ with $\zeta_x \sim \mathcal{N}(1, 1)$: the old model hits label y with probability $\sigma(b_c(y) + \gamma e_x(y))$, the new one with $\sigma(b_c(y) + \delta_c(y) + \gamma(1 + \beta)e_x(y) + u_{\text{img}})$, so δ moves each cell's operating point, β uses the evidence better and the image effect u clusters the hits. With the design fixed and labels and hits redrawn, the image-clustered case-level interval covers 96.3% and the jackknife share interval 97.3% of 400 replicates, at a true share of 0.79.

Condition (ii) is what rules out a single cluster dominating the sum – an image contributing a positive fraction of all relations, for instance – and is a standard requirement for cluster-robust inference generally, not specific to this estimator. This theorem licenses reading the image-clustered intervals reported throughout Section 4 as asymptotically valid in the idealized regime where every eligible cell's size grows without bound. That regime is not attained in-sample: VG150's eligible cells are heavy-tailed, and 2,286 of the 6,346 eligible class-pair cells hold fewer than five examples (Table 10). Those cells carry only 2.7% of identified relations, and the median eligible cell holds appreciably more than the minimum — but a third of the cells still sit where condition (iv)'s divergence is plainly hypothetical. The reported 94.0% empirical coverage against a nominal 95% (Section 4), not this theorem directly, is therefore the evidence that the interval remains a good approximation well short of the idealization.

Table 5. **Coverage of the nominal 95% case-level interval** by cell size, 400 replicates of about 4,000 relations each (`sim_small_cells.py`). SE/SD: mean estimated standard error over the standard deviation of the estimates.

Cell sizes	coverage	SE / SD
every cell $n_c = 2$	81.5%	0.71
every cell $n_c = 3$	94.5%	0.97
every cell $n_c = 4$	95.0%	0.98
every cell $n_c = 8$	96.0%	1.07
every cell $n_c = 20$	97.0%	1.09
VG150 cell sizes	96.3%	1.05

K. Choice of the grouping variable

The headline share under other groupings. We recompute the graph-constrained $mR@50$ split of TDE against the baseline (Sec. 5) with the cell declared differently (`run_sgg_grouping_share.py`); cells with one relation drop out of each version. With the class pair, 87% of the $+0.0972$ is group-level. Splitting each pair by the object's position relative to the subject (four quadrants, 9,974 cells) gives 90%, with case-level part $+0.0100$, $[+0.0065, +0.0135]$, and by the subject box's area tercile (8,230 cells) 90%, $+0.0095$, $[+0.0060, +0.0131]$: finer cells that the model could use leave the reading unchanged. Coarsening reverses it. With the subject class alone (150 cells) only 18% is group-level and the case-level part is $+0.0794$, $[+0.0730, +0.0859]$, because the object class is then a within-cell signal that the frequency prior already used: a model that moves probability according to the object's class is credited with discrimination. The share describes the declared grouping: by Proposition 3 the coarse case-level part is the fine one plus a covariance between the cells' gains and their label priors, which here is large and positive.

Fineness is not a total order, and the guidance needs that qualification. The advice to prefer the finest defensible ϕ is sound against *coarsening*, which is what Proposition 3 governs. It gives no guidance at all between two partitions of the same granularity, neither of which coarsens the other — and there the estimand can differ without limit. Four cases suffice. Take two labels, q_0 uninformative at $(\frac{1}{2}, \frac{1}{2})$, and a new model that is confidently right on all four: $q_1 = (0.9, 0.1)$ on the two cases labelled 1 and $(0.1, 0.9)$ on the two labelled 2. The total gain is $\widehat{\Delta T} = +0.480$ under every partition, and every partition below is fully identified at 100% coverage:

declared ϕ	cells	$\widehat{\Delta P}$	$\widehat{\Delta R}$
$\{1,3\},\{2,4\}$ (labels differ within each cell)	2×2	-1.120	$+1.600$
$\{1,2,3,4\}$ (one cell)	1×4	-0.587	$+1.067$
$\{1,2\},\{3,4\}$ (labels agree within each cell)	2×2	$+0.480$	0.000

The first and third partitions are equally fine, both fully identified, and they disagree about whether this model pair's improvement is entirely case-level or entirely transported. The third is the degenerate case the paper already warns about — grouping by something that determines the label leaves each cell label-homogeneous and drives $\widehat{\Delta R}$ to zero algebraically — but that warning concerns the choice $\phi = Y$, and the point here is more general and less comfortable: any ϕ correlated with the label inside a cell partially degenerates the estimand, "finest defensible" cannot detect it, and Proposition 3 is silent because neither partition is a coarsening of the other. What the analyst actually needs to declare, and what we recommend reporting alongside ϕ, is the within-cell label heterogeneity that remains: a partition whose cells are nearly label-homogeneous cannot identify much of anything, and the coverage statistic does not reveal it. This is a limitation of the construction rather than of any application of it, and we do not have a procedure that resolves it.

A recalibration-invariant alternative, and what it would cost. Because ΔR is a covariance between forecast values, its sensitivity to recalibration is structural rather than incidental, and a reader may reasonably ask why we do not use a rank-based within-cell contrast instead — a within-cell paired AUC difference, say, for which DeLong et al. [10] already supply the correlated-U-statistic variance. Such a quantity would be invariant to any monotone rescaling of either model and would dissolve the confound this paper spends a protocol on. We think the trade is worth stating and is not obviously in our favour on every axis. What a rank-based contrast gives up is the exact identity: it is not a component of the score difference, so it cannot be added to anything to recover ΔT, and a benchmark reporting it learns how the models rank cases but not how

Table 6. The paper's vocabulary, and what it corresponds to in standard terms.

Term used here	Standard counterpart	Meaning
grouping variable ϕ	stratifying / conditioning variable	a discrete function of the input, declared before the audit, whose levels the labels depend on
cell c	level of ϕ; stratum	the set of test cases sharing one value of ϕ
label transport	within-stratum relabelling; permutation within blocks	replacing a case's label with another case's label from the same cell
total gain ΔT	paired score differential	the observed mean score of the new model minus that of the old
transported gain ΔP	no exact counterpart; a mean-forecast-fit term minus a dispersion term (Proposition 1), *not* a measure of prior fit	the part of ΔT that survives label transport
within-group covariance gain ΔR	the covariance term of a covariance rearrangement of a proper score [46]; twice a classical resolution difference *only* under within-cell calibration	the part destroyed by transport; a within-cell prediction–label covariance
coverage	identified fraction	share of test cases in cells of size ≥ 2, the only cells in which ΔR is identified
calibration-matched	temperature-scaled on held-out data	both models rescaled to comparable confidence before the audit

the reported score gain decomposes. The construction here buys exact additivity with respect to a declared proper score, and pays for it with scale dependence. A rank-based diagnostic is the natural companion rather than a competitor, and reporting both — one exact and scale-dependent, one invariant and non-additive — is what we would now recommend to a benchmark that could choose.

Bounding $\sum_y \mathrm{Var}(b_y \mid c)$ label by label with Popoviciu's inequality would give $K/4$ and hence $|\rho|KM/2$, but that discards Eq. (15): the b_y are not K unconstrained quantities in $[0,1]$ but a point on the simplex, and $K/4$ is unattainable for $K > 2$. At VG150's $K = 50$ and $M = 2$ the free bound is $14.00|\rho|$ against the label-by-label $50.00|\rho|$, a factor of 3.57.

An analyst willing to bound $|\rho| \leq \bar{\rho}$ – how strongly an unrecorded variable could jointly move a model's within-cell score gain and the within-cell label distribution – obtains a numeric limit on how far $\widehat{\Delta R}_c$ can be from what a fully adjusted audit would find, without observing or modeling Z, in the spirit of Rosenbaum [32] and recent partial-correlation sensitivity models for unmeasured confounding [7]. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds or risk ratio, it can be elicited the same way across benchmarks.

Both forms of the bound are evaluated on this paper's own Visual Genome estimates in Section I. The short version is that the free bound is directionally correct but far too loose to certify anything, while the sharp per-cell form is *less* reassuring than the free one: an unrecorded covariate whose aggregate correlation with both channels reaches 0.061 would account for the entire covariance gain reported in Section N.2. We therefore do not claim those gains are robust to confounding, and Section 5's audit is unaffected only because its estimate is already indistinguishable from zero.

Section J discusses what condition (iv) costs — it forces every cell to be eventually identified, so the theorem does not describe the finite-sample regime our own applications sit in — and states the additional condition on the number of cells that the dataset-level remainder requires. What licenses reading the reported intervals at finite N is the coverage simulation of Section 4, not this theorem.

L. Terminology

M. Why the transported gain is not prior fit

Proposition 1 splits ΔP_c into two brackets, and only the first is prior fit: it is positive exactly when the new model's average within-cell forecast sits closer to the cell's label frequencies. The second has nothing to do with the prior. It rewards the new model for making *less* dispersed predictions inside the cell, at a fixed mean forecast, so ΔP can be large and positive when neither model's prior fit has changed at all. One cell suffices: with two labels and frequencies $(\frac{1}{2}, \frac{1}{2})$, let q_0 alternate between $(1,0)$ and $(0,1)$ in a pattern uncorrelated with the label and let q_1 predict $(\frac{1}{2}, \frac{1}{2})$ throughout. Both mean forecasts equal the cell frequencies exactly, so the first bracket is zero, and the whole observed gain of $\frac{1}{2}$ is transported, produced

Table 7. Frozen-model top-1 accuracy on VG150 PredCls. Similar accuracies do not reveal whether the difference is transported.

Model	Accuracy
FREQ	0.6564
FREQ+OVERLAP	0.6545
MLP-CLASS	0.6554
MLP-SPATIAL	0.6639
MLP-SPATIAL+	0.6674

entirely by the collapse in dispersion. The channel is therefore the reliability-and-sharpness channel of the classical partition, not a prior-fitting channel, which is why we name it for the operation. Two consequences: a claim that some gain is "mostly prior-fitting" does not follow from a large $\widehat{\Delta P}$ alone, and Eq. (8) explains why confidence matching moves $\widehat{\Delta P}$ so sharply, since a temperature change is almost purely a change in dispersion.

N. Further empirical studies

This section reports the two studies summarized in Section 6 and the granularity, score and cell-size sensitivity analyses referred to there. Each applies the estimator of Section 3 without modification; only the declared grouping variable ϕ changes.

N.1. Controlled predictors on VG150 in full

We work with Visual Genome [20] in the PredCls regime [44], where ground-truth boxes and object labels are supplied and the model has only to predict one of 50 predicate labels for each ordered object pair. The predictors of this section are audited on a VG150-*style* dataset we build directly from the released relationship annotations: the 150 most frequent object classes and 50 most frequent predicates form the vocabulary, and images are split $70/30$ into a pool for fitting the compared predictors and a pool on which the audit runs. We use this reconstruction rather than the canonical VG150 release because these predictors are trained here from annotations alone; Section 5 audits *released third-party checkpoints* on the canonical split itself, so both settings appear in the paper and neither is asked to stand for the other. The reconstruction provides 532,987 training and 229,605 test relations, drawn from 65,228 and 27,955 images respectively, and the cell throughout is $\phi = (\text{subject class}, \text{object class})$, two real examples of which appear in Figure 5. Of the 8,614 cells present in the test split, 6,346 contain at least two relations and are therefore identified; the remaining 2,268 are singletons, which the estimator excludes and reports rather than smoothing over, leaving 227,337 relations (99.0%) under audit.

The five predictors are deliberately constructed so that each exposes a different kind of information. FREQ is the training-count distribution conditioned on the class pair, with add-one smoothing and a subject-marginal and then global backoff for class pairs unseen in training; FREQ+OVERLAP adds to it a single binary box-overlap indicator. MLP-CLASS consumes fixed subject and object class embeddings, which makes it a function of ϕ alone; MLP-SPATIAL extends those embeddings with 12 normalized box-geometry features covering relative centre, scale, aspect, intersection, containment and IoU, while MLP-SPATIAL+ both widens the network and trains longer (Section H lists every architecture and schedule), so its increment over MLP-SPATIAL reflects capacity and optimization jointly rather than capacity alone. Each is trained once on the training split, and the split thereafter consumes nothing but their cached test distributions, scored quadratically with image-clustered intervals and 499 within-cell cyclic randomizations, whose smallest attainable p-value is $1/500 = .002$; table entries of .002 are at this granularity floor and should be read as $p \leq .002$. With confidence intervals as the primary uncertainty statement and every significant effect many standard errors from zero, we do not additionally adjust the diagnostic p-values for the number of reported comparisons.

N.2. Gain attribution for the controlled predictors

The central result is collected in Table 8 and Figure 6. MLP-CLASS improves on FREQ by 0.03386 of quadratic score, yet because its gain vector is constant inside every class-pair cell, the estimator assigns that entire improvement to prior transport and precisely nothing to instance covariance. The zero is worth dwelling on: it follows algebraically from the construction rather than from a learned baseline happening to pass a sanity check.

Adding box geometry changes the picture. MLP-SPATIAL gains 0.04270 over FREQ, of which 0.02832 survives label transport and 0.01439 is within-group covariance (CI $[0.01373, 0.01504]$). More revealing still is the direct comparison

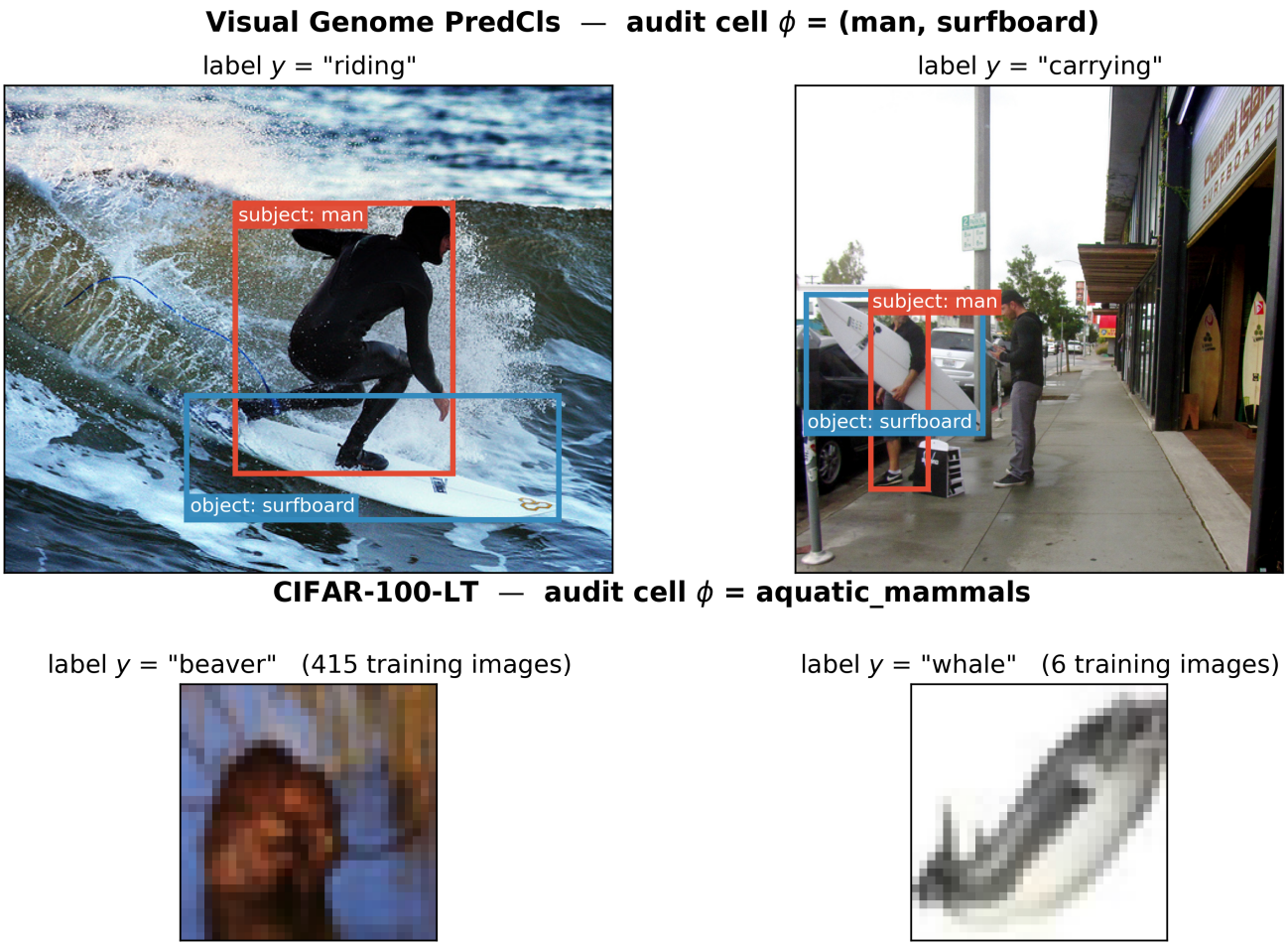


Figure 5. **What a cell contains, on both benchmarks.** Each row shows two real test examples that share the same declared grouping variable ϕ but carry different labels – precisely the configuration within-cell transport exploits. *Top:* two Visual Genome relations from the cell $\phi = (\text{man}, \text{surfboard})$. The class pair is identical and therefore cannot distinguish "riding" from "carrying"; only the image content can, which is what ΔR is designed to credit. Boxes are the annotated subject (red) and object (blue) regions that MLP-SPATIAL turns into geometry features and MLP-VISUAL crops and encodes with CLIP. *Bottom:* two CIFAR-100 test images from the superclass $\phi =$ aquatic_mammals, drawn from opposite ends of the long-tailed training distribution (415 versus 6 training images). Crossing the labels within a row preserves the cell and its label histogram while destroying which prediction belongs with which instance.

against MLP-CLASS, where the total gain falls to 0.00885 only because the transported channel deteriorates by 0.00554 even as the covariance gain from spatial features rises by 0.01439. The resulting covariance share of 1.63 records a genuinely useful instance-specific change that the aggregate score partly conceals behind a worse prior component, and it is exactly the configuration that a ratio confined to $[0, 1]$, or thresholded at 0.5, would misdescribe.

The remaining two comparisons bracket the interesting range. MLP-SPATIAL+ adds 0.00374 over MLP-SPATIAL, nearly all of it covariance gain at 0.00321 (CI $[0.00289, 0.00353]$), whereas FREQ+OVERLAP supplies the opposite caution: its overlap indicator does create a small positive covariance gain, but the accompanying negative prior term is larger, so the model ends up worse overall. Using image-derived information is thus never treated as an improvement in itself; both channels are reported, signs included.

Reading the magnitudes. Quadratic-score units are not ones most readers calibrate by eye, so it is worth anchoring them in the rank-based metrics this literature reports. Where both channels point the same way the correspondence is straightforward: MLP-SPATIAL-S's $\widehat{\Delta R} = +0.01023$ over MLP-CLASS-S accompanies $+0.51$ points of top-1 predicate accuracy, $+0.60$ of mean reciprocal rank and $+0.86$ of recall@5, so in these data a hundredth of quadratic-score gain is worth roughly half an

Table 8. The split on 227,337 identified VG150 PredCls relations. Gains use the quadratic score. CI is image-cluster robust; p is the 499-shift relation-level randomization diagnostic (granularity floor .002). "–" denotes an undefined share because total gain is negative. The MLP-CLASS row's zero-width interval is exact by construction, not a sampling statement: a gain vector constant within every cell makes the antisymmetric kernel identically zero (Theorem 1), so no variability exists for the interval to measure.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	$\widehat{\rho}$	p
FREQ+OVERLAP	FREQ	-0.01229	-0.01430	$0.00202\,[0.00168, 0.00235]$	–	.002
MLP-CLASS	FREQ	0.03386	0.03386	$0.00000\,[0.00000, 0.00000]$	0.00	.724
MLP-SPATIAL	FREQ	0.04270	0.02832	$0.01439\,[0.01373, 0.01504]$	0.34	.002
MLP-SPATIAL+	FREQ	0.04644	0.02885	$0.01760\,[0.01686, 0.01834]$	0.38	.002
MLP-SPATIAL	MLP-CLASS	0.00885	-0.00554	$0.01439\,[0.01373, 0.01504]$	1.63	.002
MLP-SPATIAL+	MLP-SPATIAL	0.00374	0.00053	$0.00321\,[0.00289, 0.00353]$	0.86	.002

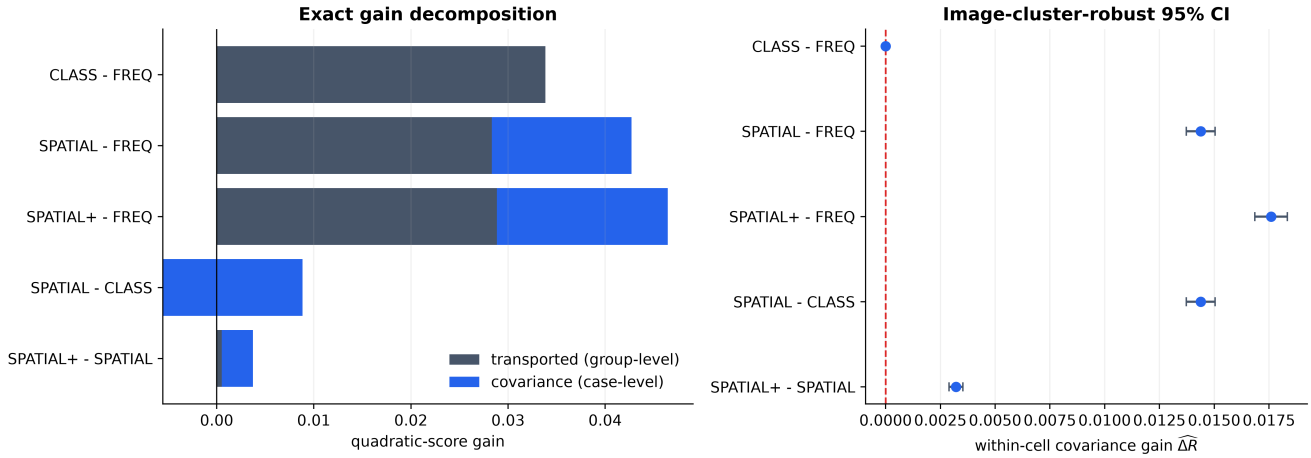


Figure 6. **Direct model-pair attribution.** Left: exact decomposition of each positive gain into transported and within-group covariance components (the negative prior term for SPATIAL–CLASS extends left of zero). Right: primary covariance estimates with image-cluster-robust intervals.

accuracy point. The covariance channel is not, however, a re-description of any rank metric, and the real-pixel comparison of Section 6 is where that becomes visible: there $\widehat{\Delta R} = +0.00640$ while top-1 accuracy falls by 3.11 points and recall@5 by 2.20. Prior-matching both models before scoring narrows the accuracy gap to 2.17 points without closing it. The rank metrics aggregate the two channels into one number; when the channels carry opposite signs, as they do here, no single such number can represent both, which is precisely the situation the decomposition exists to report.

Reading $\widehat{\Delta T}$ as a comparative forecast test. Corollary 3 identifies $\widehat{\Delta T}$ and its image-clustered interval in Table 8 as a grouped Diebold–Mariano statistic and interval, on the identified subsample, for each pair's paired score differential, so every row already reports what that classical test would report. What it does not report is the next two columns: whether MOTIFS-TDE (Section 5) or MLP-SPATIAL beat their baseline for a reason a frequency-only remodeling of the same benchmark could not reproduce, which is the question $\widehat{\Delta P}$ and $\widehat{\Delta R}$ answer and DM/GW do not ask.

N.3. Real pixels against a geometric proxy, in full

None of the predictors above ever sees an image: they are functions of the class pair and, for MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry is only a proxy for image content, a sceptical reader may ask whether the covariance channel responds to pixel evidence or merely to that proxy. Figure 5 sharpens the question — the cell $\phi = (\text{man}, \text{surfboard})$ holds both "riding" and "carrying", and what distinguishes them is what the photograph shows.

MLP-VISUAL answers it. Subject and object regions are cropped from the original image and each encoded by a frozen

Table 9. Matched-subsample real-pixel study on the full VG150 test split (coverage 99.0%, as in Table 8). All three predictors are trained on the same 100,000 relations and differ only in features. CI is image-cluster robust on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
MLP-SPATIAL-S	MLP-CLASS-S	0.00503	−0.00520	0.01023 [0.00967, 0.01079]	.002
MLP-VISUAL-S	MLP-CLASS-S	−0.04139	−0.05802	0.01663 [0.01533, 0.01794]	.002
MLP-VISUAL-S	MLP-SPATIAL-S	−0.04642	−0.05282	0.00640 [0.00513, 0.00767]	.002

CLIP ViT-B/32 encoder [31]; the two 512-dimensional embeddings are concatenated with the class embeddings MLP-CLASS already uses and passed to a small trained head. Because CLIP is never fine-tuned, whatever covariance the model gains reflects information its pretraining already recovered from pixels rather than representation learning on Visual Genome.

Encoding every training relation with CLIP is expensive, so MLP-VISUAL must be trained on a subsample; setting a subsampled model against the full-data MLP-SPATIAL would confound feature content with training-set size. All three compared predictors are therefore retrained on one fixed, seeded subsample of 100,000 of the 532,987 training relations, written MLP-CLASS-S, MLP-SPATIAL-S and MLP-VISUAL-S, so any gap between them isolates the features. The audit still uses the full 229,605-relation test split with the same class-pair ϕ and image-clustered intervals.

Result. By any aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6156 against 0.6467 for MLP-SPATIAL-S and 0.6416 for MLP-CLASS-S, and a quadratic score 0.04642 below MLP-SPATIAL-S. An evaluator reading either number would discard it.

The decomposition contradicts that reading on the axis that matters. Against MLP-SPATIAL-S the CLIP model loses 0.05282 of transported gain while *gaining* 0.00640 of covariance (95% CI $[0.00513, 0.00767]$, randomization $p = .002$): its entire deficit sits in the transported channel and its within-cell discrimination is significantly better, not worse. Against the class-only baseline the point is plainer still — pixels buy 0.01663 of covariance where box geometry buys 0.01023, with well-separated intervals. Real image content carries more case-specific signal than the geometric proxy standing in for it.

A plausible mechanism, stated as hypothesis rather than finding: the 1024 CLIP dimensions dominate the 64 dimensions of class embedding the head also receives, so the model fits VG150's class-pair frequency structure markedly less well than a predictor built around it. A benchmark whose headline metric rewards transported gain will punish this model, and the aggregate numbers duly do. What the decomposition adds is that the punishment is not for failing to use the image. Two falsification checks, reported in Section H, support that reading: correcting a model's within-cell prior towards the training histogram registers as almost pure transported movement (+0.02390 against +0.00114 of covariance), as a correction carrying no case-level information must, and applied to the CLIP model it halves the aggregate deficit (−0.04642 to −0.02138) while the covariance advantage persists (+0.00754, and +0.00664 when both models are corrected) and survives confidence matching (+0.00464, CI [+0.00311, +0.00618]).

N.4. Sensitivity analyses

For MLP-SPATIAL versus MLP-CLASS, the conclusion survives every planned sensitivity check (Table 10). Changing from the quadratic to a probability-floored log score changes scale (0.01439 to 0.06536) but not sign or significance. The coverage column matters for reading the rows against each other, and earlier versions of this table omitted it. Each row targets the identified subpopulation its own setting defines, so the minimum-cell-size rows are not four estimates of one quantity: raising the minimum from 2 to 20 drops coverage from 99.0% to 86.7% and changes the estimand along with the estimator. That the estimate moves so little across them (0.01439 to 0.01225) is therefore mild evidence of stability rather than a sensitivity analysis holding the target fixed.

Coarsening ϕ from the class pair to subject class increases the credited covariance to 0.02181. Proposition 3 identifies why: the coarser cell's estimate equals the class-pair estimate's conditional average plus a between-class-pair covariance term, and that term is positive here, meaning some of the credited gain reflects class-pair-level structure that the finer audit had already correctly assigned to ΔP. This is precisely the sense in which the class-pair audit is the conservative choice: it is not that coarsening is wrong, but that it cannot be assumed neutral, and only the finer partition's estimate isolates covariance gain net of every prior regularity expressible in ϕ. The same class-pair-versus-subject-only pair also instantiates Proposition 4 directly, with the subject class playing the role of the unrecorded confounder Z: Table 4 in Section I reports that the resulting bias of 0.00742 is comfortably inside the Cauchy–Schwarz bound, though the bound is loose enough at $K = 50$ that it would

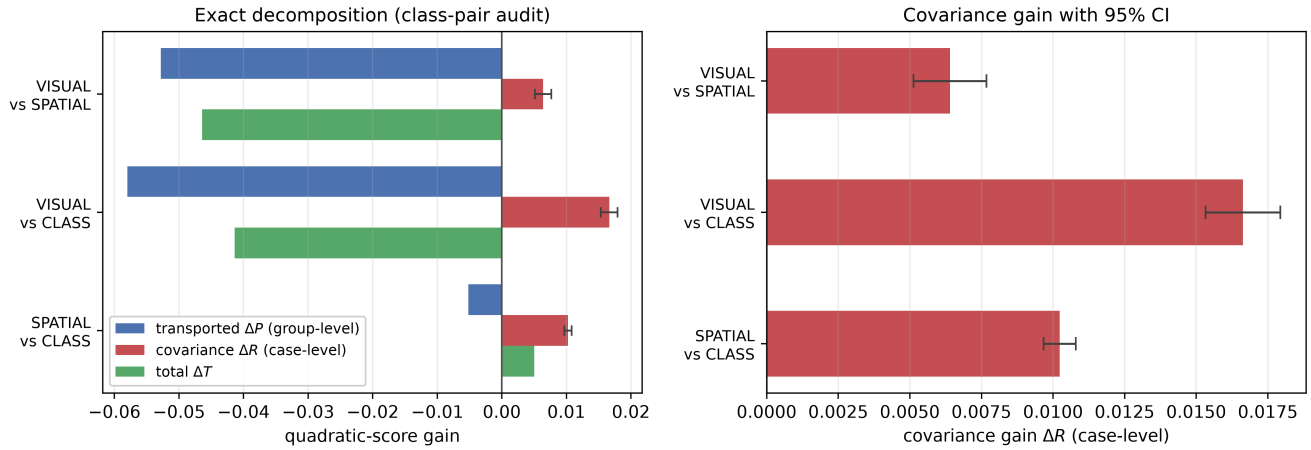


Figure 7. **Pixels beat geometry on covariance while losing on the aggregate.** Left: the exact decomposition for the three matched-subsample pairs. Both comparisons involving the CLIP model have strongly negative totals (green) driven entirely by the transported channel (blue), while their covariance channel (red) is positive in every case. Right: the covariance channel alone, with 95% image-clustered intervals. Replacing box geometry with real image content raises within-group covariance from 0.01023 to 0.01663 over the same class-only baseline.

Table 10. Sensitivity of the MLP-SPATIAL vs. MLP-CLASS decomposition to score, audit granularity, and minimum cell size. CI is image-cluster robust on $\widehat{\Delta R}$.

Factor	Setting	$\widehat{\Delta T}$	$\widehat{\Delta R}$ (95% CI)	Cells	Cov.
Score	Brier (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Score	Log (floored)	0.04297	0.06536 [0.06294, 0.06779]	6346	99.0%
Cell ϕ	class pair (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Cell ϕ	subject only	0.00943	0.02181 [0.02079, 0.02283]	150	100.0%
Min. cell size	2 (def.)	0.00885	0.01439 [0.01373, 0.01504]	6346	99.0%
Min. cell size	5	0.00772	0.01395 [0.01329, 0.01462]	4060	96.3%
Min. cell size	10	0.00683	0.01330 [0.01262, 0.01398]	2744	92.6%
Min. cell size	20	0.00570	0.01225 [0.01155, 0.01295]	1757	86.7%

remain satisfied for a $\bar{\rho}$ three orders of magnitude smaller than one. Raising the minimum cell size from 2 to 5, 10, and 20 yields 0.01395, 0.01330, and 0.01225; all cluster-robust lower limits remain positive. The procedure itself has no smoothing strength, split fraction, or decision threshold to tune.

N.5. Long-tailed image classification

Sections 4–6 are all instances of one task (predicate classification on an ordered object pair). Section 2 argued that long-tailed image classification poses a structurally identical question: a class-frequency prior can be fit better without any per-instance improvement. We test this directly on CIFAR-100-LT [8], an exponential-imbalance (ratio 100) subsample of CIFAR-100's training split with the original, balanced test split retained for evaluation. We train ResNet-32 [17] classifiers that differ only in their per-class loss weighting: a vanilla cross-entropy baseline (CE); class-balanced effective-number re-weighting applied from initialization (CB) [8]; and the same re-weighting deferred to the final twenty epochs (DRW) [3]. Optimizer, schedule, augmentation, epoch budget, and seed are identical across runs (Section H).

Choosing ϕ. A declared grouping variable has to predict the label without determining it, and this second benchmark makes the constraint unusually easy to violate. Were ϕ taken to be the fine class label itself, every cell would be label-homogeneous, within-cell transport would have nothing to cross, and $\widehat{\Delta R}$ would come out identically zero – an algebraic artifact wearing the appearance of a finding. CIFAR-100's own 20 coarse superclasses avoid this, each gathering five fine labels into a cell of 500 balanced test images, which is structurally the situation of a VG class-pair cell holding several predicates (Figure 5). Unlike a

class pair, though, a superclass is a function of the label, not of the input, and no model observes it at test time: the superclass rows show how the estimator behaves under a label-derived grouping, not an audit of a feature both models condition on. The input-grounded row is the global cell, which every model trivially observes. Alongside the superclass audit we report two further declared priors: the trivial coarsening to a single global cell, which audits against the global class marginal and so instantiates Proposition 3 in a second domain, and a partition into head, medium and tail training-frequency tiers. CIFAR-100 test images are independent, unlike the several relations VG draws from one photograph, so no cluster grouping is required and the plain Hájek interval of Section 3 applies directly. One semantic point matters in this domain: transport uses each cell's *test* label histogram, which on CIFAR-100-LT is balanced. A positive ΔP here therefore records movement *toward* the balanced test marginal – undoing the training-frequency skew – which is precisely the adjustment long-tailed evaluation is designed to reward; a transported gain is a description of where score was earned, not an accusation (Section 7).

Result. The three classifiers reach top-1 accuracies of 0.3662 (CE), 0.2627 (CB) and 0.3874 (DRW), so re-weighting from initialization degrades the classifier substantially while the deferred schedule delivers the improvement it was designed to produce. Table 13 decomposes both gains against the superclass audit.

Of the two, the class-balanced run is the more instructive, precisely because its aggregate score conceals almost everything of interest. On the quadratic score it gains $+0.00143$ over the baseline, a difference small enough that a reader working from the headline number alone would reasonably conclude that re-weighting had changed very little. The decomposition tells another story: within superclasses the transported channel improves by $+0.19713$ while within-group covariance deteriorates by -0.19569 (95% CI $[-0.20728, -0.18411]$), so the near-zero total is not a small effect but two large ones that happen to cancel.

A calibration-matched regime. A channel split of that shape could, however, be produced by confidence scaling alone. Under the quadratic score the covariance channel is a covariance between probability movements and labels (Eq. (3)), and a monotone recalibration moves it: softening an overconfident model transfers score mass from the covariance channel to the transported channel while adding no information about any individual image. The baseline invites exactly this concern, assigning a mean maximum probability of 0.764 while being correct on only 0.3662 of the test set. We therefore repeat the audit in a calibration-matched regime: each model's temperature is chosen by held-out likelihood on a calibration half of the test split ($T^* = 2.85$ for CE, 3.51 for CB, 2.61 for DRW), the decomposition is computed on the disjoint audit half, and the whole procedure is repeated over twenty random splits (Table 11). The regime's necessity is quantified by its control row: recalibrating the baseline alone – a transformation carrying no instance information – moves the channels by $+0.380$ (prior) and -0.175 (covariance), so the raw split does mix calibration with discrimination.

The calibration-matched rows then separate the two schedules cleanly, and more sharply than the raw ones. CB's covariance deficit does not shrink but widens, to $-0.23991\,(0.00491)$: the model re-weighted from initialization has genuinely lost within-superclass discrimination – a real degradation that the ten-point accuracy drop records independently – and its near-zero raw total reflects that loss offset by its softer, better-calibrated confidence. DRW inverts the reading its raw split suggests. Raw, its $+0.06712$ gain divides roughly two-thirds transported to one-third covariance gain; calibration-matched, the improvement is $+0.02173\,(0.00275)$ of which $+0.01957\,(0.00329)$ is within-group covariance while the transported channel is close to zero ($+0.00216\,(0.00410)$). A caveat applies, in the light of Sec. 5: matching here is by temperature only, which cannot remove a class-wise shift of the operating point, and the post-hoc logit-adjusted baseline's matched covariance, $+0.03241$ (Tab. 12), exceeds DRW's, $+0.01953$; part of any such covariance may therefore be an operating-point effect, and the groups are label-derived, so "group-level" describes the label partition, not the input. What survives calibration matching is precisely the component the deferred schedule was designed to buy – per-instance discrimination, concentrated on rare classes (Table 14) – while the apparently transported share of its raw gain was largely the calibration difference between the two runs. Neither an accuracy difference nor an aggregate score exposes either composition, which is the practical argument for reporting the two channels, in both regimes, instead of their sum.

Seeds, imbalance ratios, and a post-hoc arm. A decomposition computed from one trained pair carries only test-set sampling uncertainty, which is not the uncertainty relevant to a claim about a training *method*. We therefore retrained the triple at two further seeds at the primary ratio and at imbalance ratios 10 and 50, and added a fourth arm: post-hoc logit adjustment (LA), which divides the baseline's predictions by the training class prior without retraining [28]. Every configuration is audited in both regimes; Table 12 reports the calibration-matched covariance channel, which is the one the preceding paragraphs interpret.

Table 11. Raw and calibration-matched decompositions on CIFAR-100-LT at the superclass audit. Temperatures are fitted by likelihood on a held-out calibration half of the test split; every decomposition uses the disjoint audit half; entries are means (sd) over twenty random splits. The final row is the control: recalibrating the baseline alone carries no instance information yet moves both channels substantially.

Comparison	Regime	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$
CB vs CE	raw	$+0.00208\,(0.00642)$	$+0.19635\,(0.00369)$	$-0.19427\,(0.00632)$
CB vs CE	calibration-matched	$-0.12912\,(0.00373)$	$+0.11079\,(0.00372)$	$-0.23991\,(0.00491)$
DRW vs CE	raw	$+0.06712\,(0.00865)$	$+0.04258\,(0.00463)$	$+0.02454\,(0.00685)$
DRW vs CE	calibration-matched	$+0.02173\,(0.00275)$	$+0.00216\,(0.00410)$	$+0.01957\,(0.00329)$
CE recalibrated vs CE	control	$+0.20495\,(0.00420)$	$+0.37960\,(0.00268)$	$-0.17464\,(0.00455)$

Table 12. Calibration-matched covariance gain across seeds and imbalance ratios, with balanced test accuracy for each arm. LA is post-hoc logit adjustment applied to the baseline. The class-balanced deficit scales with imbalance severity; the deferred schedule's gain is positive throughout but seed-sensitive in magnitude; the post-hoc correction lands in the covariance channel rather than the transported channel.

		Top-1 accuracy				$\widehat{\Delta R}$ vs. CE (calibration-matched)		
Ratio	Seed	CE	CB	DRW	LA	CB	DRW	LA
10	0	0.5388	0.5154	0.5447	0.5497	-0.06903	$+0.00617$	$+0.02222$
50	0	0.4256	0.3242	0.4399	0.4550	-0.24409	$+0.01903$	$+0.03768$
100	0	0.3662	0.2627	0.3874	0.3959	-0.23936	$+0.01987$	$+0.03449$
100	1	0.3614	0.2287	0.3762	0.3877	-0.26176	$+0.00748$	$+0.02672$
100	2	0.3687	0.2481	0.3968	0.4024	-0.25213	$+0.03124$	$+0.03601$
100, mean (sd)						$-0.25108\,(0.01124)$	$+0.01953\,(0.01188)$	$+0.03241\,(0.00498)$

Three things follow. First, the class-balanced run's covariance loss is not a misconfiguration but a systematic function of imbalance severity: -0.069 at ratio 10, -0.244 at 50, and $-0.251\,(0.011)$ across three seeds at 100, tracking an accuracy penalty that grows from 2.3 to 10.4 points over the same range. Second, the deferred schedule's covariance gain is positive at every ratio and in every seed, but its magnitude varies roughly fourfold across seeds at ratio 100 ($+0.007$ to $+0.031$). We therefore report the spread rather than a seed-level interval: the direction replicates, the magnitude is seed-sensitive, and the gap between the tight within-run intervals of Table 13 and this across-seed spread is exactly why a method-level claim needs replication rather than a single audit.

Third, LA is an instructive falsification of a natural expectation. A purely post-hoc prior correction might be expected to register as almost pure transported channel, as the within-cell prior matching of Section 6 does. It does not: calibration-matched, LA reads as almost entirely covariance gain ($+0.022$ to $+0.038$) with a transported channel near zero, while achieving the best balanced accuracy of any arm at every ratio. The reason is visible in the construction. Dividing by the *global* class prior is not a within-superclass histogram move, because fine classes inside a superclass differ in training frequency, and the correction it applies is multiplicative in each example's own predicted probabilities — it changes rankings where the model already carried latent evidence. The two post-hoc corrections in this paper therefore land in opposite channels, and correctly so: transport separates adjustments that act on the cell's label histogram from adjustments that act on individual predictions.

Where the covariance gain sits is visible in Table 14 and Figure 8 (raw regime, primary seed). For DRW it falls where the long-tail literature intends it to, reaching $+0.09009$ on few-shot classes and $+0.06760$ on medium-shot ones, while many-shot classes lose -0.07624 as the schedule shifts capacity away from the head. The class-balanced run reproduces that same ordering with every level displaced downward, its many-shot covariance collapsing to -0.41977. What the decomposition supplies, then, is not only how much of a long-tail gain is real but where in the frequency spectrum it was earned.

These numbers also provide an implementation check on real inputs. Computing the within-group covariance gain directly from the identity of Eq. (3), through a code path that shares nothing with the transport estimator, gives -0.19530; multiplying by the factor $n_c/(n_c-1)$ that Proposition 5 predicts, at the $n_c = 500$ of a balanced superclass cell, returns -0.195693 against the estimator's -0.19569. The identities of Theorem 1 and Proposition 5 are algebraic, so this verifies the implementation rather than the theory; the point of the check is that two independent code paths agree to the fifth decimal on real trained

Table 13. The split on the 10,000-image balanced CIFAR-100-LT test split. Gains use the quadratic score; CI is the Hájek interval on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic, one-sided for a positive covariance gain, so the value of 1.000 on the CB rows records a covariance gain at the extreme negative end of its randomization distribution.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (20)	0.00143	0.19713	$-0.19569\,[-0.20728, -0.18411]$	1.000
CB vs CE	global (1)	0.00143	0.25573	$-0.25430\,[-0.26801, -0.24058]$	1.000
CB vs CE	tier (3)	0.00143	0.25247	$-0.25103\,[-0.26453, -0.23754]$	1.000
DRW vs CE	superclass (20)	0.06606	0.04206	$0.02400\,[0.01340, 0.03460]$	.002
DRW vs CE	global (1)	0.06606	0.03522	$0.03085\,[0.01860, 0.04309]$	.002
DRW vs CE	tier (3)	0.06606	0.03702	$0.02904\,[0.01702, 0.04107]$	.002

Table 14. Covariance gain by training-frequency tier at the primary superclass audit. The deferred schedule concentrates its genuine per-instance improvement on rare classes; both methods lose covariance on the head.

		CB vs CE	DRW vs CE	
Tier	n	$\widehat{\Delta R}$	$\widehat{\Delta T}$	$\widehat{\Delta R}$
Few-shot (< 20)	3000	-0.03337	0.13785	0.09009
Medium-shot (20–100)	3500	-0.11075	0.10799	0.06760
Many-shot (> 100)	3500	-0.41977	-0.03740	-0.07624

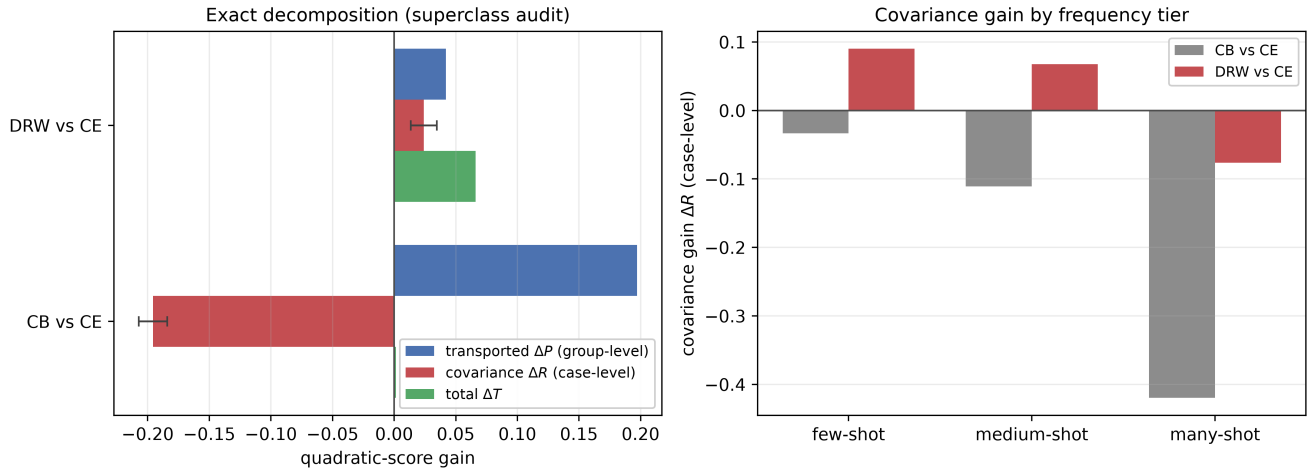


Figure 8. **The same attribution transfers to a second task.** Left: exact decomposition at the superclass audit. The CB total gain (green) is visually almost absent at $+0.00143$ while its two channels reach ± 0.2 – a near-zero aggregate score concealing two large opposing effects. DRW's smaller total is genuinely split between both channels. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: the covariance channel by training-frequency tier, showing that DRW's genuine per-instance gain is concentrated on rare classes and negative on the head.

models, not only on the synthetic populations used in the unit tests.

Coarsening behaves as Proposition 3 predicts. Moving from the 20 superclass cells to a single global cell changes the credited covariance from -0.19569 to -0.25430; the difference is the between-cell covariance term, which is negative here, so the coarse audit overstates the covariance loss by attributing to it structure that the superclass partition correctly assigns to ΔP. The finer partition is again the conservative reading, for the same reason as in Section N.4 and now in a second domain.

Table 15. The split on the 7,532-document balanced 20-Newsgroups-LT test split. Gains use the quadratic score; CI is the (unclustered) Hájek interval on $\widehat{\Delta R}$, documents being independent units; p is the 499-shift randomization diagnostic, one-sided for a positive covariance gain.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (6)	0.11388	0.07071	$0.04317\,[0.03664, 0.04970]$	.002
CB vs CE	global (1)	0.11388	0.07203	$0.04185\,[0.03381, 0.04990]$	.002
CB vs CE	tier (3)	0.11388	0.07810	$0.03578\,[0.02825, 0.04331]$	.002
DRW vs CE	superclass (6)	0.11490	0.11335	$0.00155\,[-0.00296, 0.00605]$	.236
DRW vs CE	global (1)	0.11490	0.11809	$-0.00319\,[-0.00884, 0.00245]$	.980
DRW vs CE	tier (3)	0.11490	0.12391	$-0.00901\,[-0.01428, -0.00374]$	1.000

N.6. Long-tailed text classification

Sections N.2–N.5 test two vision tasks that share cached probability vectors and grouping variables grounded in class frequency or object geometry. To test the breadth Section 7 claims – any benchmark supplying two models' full probability vectors, labels, and a declared discrete grouping variable – we apply the identical estimator, unmodified, to document category prediction on 20 Newsgroups [21]. Neither the predictors nor the grouping variable are visual here: documents replace images, and the declared ϕ is a topical grouping rather than a spatial or class-frequency one, so a favorable result cannot be attributed to anything specific to vision data.

We build a long-tailed training subset with the same exponential profile as CIFAR-100-LT (imbalance ratio 100, Section N.5), applied to 20 Newsgroups' 20 fine categories, keeping the standard balanced test split (7,532 documents) untouched (Section H). The 20 categories fall into six conventional coarse topics – computing, recreation, science, politics, religion, and classifieds – which we use as the primary declared ϕ, the direct analogue of CIFAR's coarse superclasses and, like them, a function of the label rather than of the input; the global (trivial) cell, the input-grounded row, and a training-frequency-tier partition are reported alongside, as in Section N.5. Documents are represented by a TF-IDF vector reduced to 300 dimensions by truncated SVD, and the three compared classifiers – CE, CB, and DRW – follow the same three-arm protocol as Section N.5: an identical small MLP architecture and optimizer across runs, differing only in per-example loss weight (Section H). Test documents are independent units, so the plain Hájek interval of Section 3 applies directly, as for CIFAR.

Result. Table 15 decomposes both re-weighting schedules' gain over the cross-entropy baseline (top-1 accuracies 0.3532 CE, 0.3960 CB, 0.3776 DRW). The two schedules land in opposite channels here, which is itself the finding worth reporting: on this task it is class-balanced re-weighting from initialization, not the deferred schedule, that buys genuine within-topic covariance gain ($+0.04317$, CI $[0.03664, 0.04970]$, 37.9% of its total gain), while DRW's comparably sized total gain (0.11490 vs. CB's 0.11388) is 98.6% transported, its residual covariance statistically indistinguishable from zero at the superclass audit ($+0.00155$, CI $[-0.00296, 0.00605]$, $p = .236$) and significantly negative once the coarser tier partition is used instead (-0.00901, CI $[-0.01428, -0.00374]$). This is the reverse of Section N.5's image-classification finding, where CB's near-zero total masked a large covariance-gain *loss* and DRW's gain was genuinely covariance-driven. Nothing about the estimator or the construction of ϕ changed between the two applications; what changed is which training recipe, on which data, buys real per-instance discrimination – exactly the composition question an aggregate score or an accuracy difference alone cannot answer, now demonstrated on a domain with neither a class-frequency table nor a spatial prior.

Coarsening moves both comparisons in the direction Proposition 3 allows without requiring: CB's credited covariance falls from 0.04317 at the superclass partition to 0.04185 at the global cell and 0.03578 at the tier partition, while DRW's changes sign twice, from a small positive residual (superclass) to a small negative one (global) to a significant negative one (tier). None of these differences overturns CB's conclusion, but DRW's illustrates exactly why the finest defensible partition is the conservative default (Section 3): a reader stopping at the coarser tier partition would report a significant covariance *loss* for a method whose finer-grained residual is not distinguishable from zero.

Table 16 breaks the superclass-audit covariance down by training-frequency tier for both schedules; the two methods

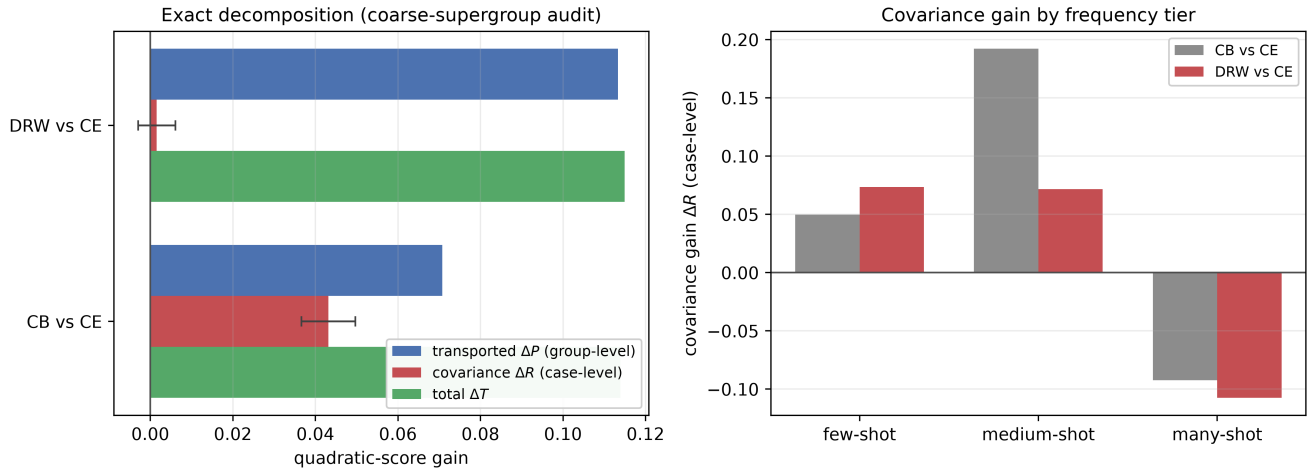


Figure 9. **A third domain, with neither a spatial nor a class-frequency prior.** Left: exact decomposition at the coarse-topic audit; CB's total gain is genuinely split between both channels while DRW's is almost entirely transported. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: covariance gain by training-frequency tier for both schedules, showing the same head-loses/tail-gains signature as Figure 8 despite the change of domain and grouping variable.

Table 16. Covariance gain by training-frequency tier at the superclass audit, 20-Newsgroups-LT. Both schedules gain within-group covariance on rare and medium categories and lose it on frequent ones.

Tier	n	$\widehat{\Delta R}$, CB vs CE	$\widehat{\Delta R}$, DRW vs CE
Few-shot (<20)	1954	0.04991	0.07352
Medium-shot (20–100)	2610	0.19213	0.07171
Many-shot (>100)	2968	−0.09227	−0.10753

agree in shape even though their superclass totals diverge sharply in composition, each gaining covariance on rare and medium-frequency categories and losing it on frequent ones, the same qualitative signature Section N.5 reports for image classification.

This third domain uses a single trained pair per schedule rather than the multi-seed, calibration-matched protocol Section N.5 applies to CIFAR-100-LT; extending that fuller protocol to text is the natural next replication, not a step this section claims to have taken.

O. Splitting mean recall on the released checkpoint, in full

One pass, every model. In PredCls the predictor's object probabilities are exact one-hot vectors, so the codebase's other two counterfactual effects are not new models: the Total Effect equals TDE, and the Natural Indirect Effect is identically zero. We therefore ran the released causal MOTIFS checkpoint once, with `EFFECT_TYPE TDE`, and patched the predictor to write, for every ordered object pair of every test image, the visual logits, the context logits, the context logits under the averaged context, the frequency logits, and the logits it returned. The returned logits equal $(\mathrm{vis}+\mathrm{ctx})+\mathrm{frq}-[(\mathrm{vis}+\mathrm{ctx}_{\mathrm{avg}})+\mathrm{frq}]$ with a largest absolute difference of exactly 0 over all pairs, which is the cancellation described in Sec. 5. The baseline is $(\mathrm{vis}+\mathrm{ctx})+\mathrm{frq}$; single branches and branch pairs are their own sums; the logit-adjusted baseline (LA_τ) subtracts $\tau \log \pi(y)$, with π the predicate frequencies of the training split (439,063 relations).

Replaying the evaluator. For model m, relation $i=(s,o,y_i)$ and any predicate y, let $r_i^{(m)}(y)$ be the position at which the official evaluator would first match the triplet (s,o,y). Under the graph constraint each pair contributes its top non-background predicate, pairs are ordered by that predicate's probability, and a predicted pair matches (s,o) when both objects carry the same classes and both boxes overlap with IoU at least 0.5; so $r_i^{(m)}(y)$ is the best rank among matching pairs whose top predicate is y. Without the graph constraint every (pair, predicate) is ranked by its probability and $r_i^{(m)}(y)$ is the best

position of (p, y) over matching pairs p. Relation i is a recall@k hit under label y iff $r_i^{(m)}(y) < k$, which makes recall an exact function of the labelling. Matching through duplicate objects matters: a relation can be recalled by a different pair of near-identical boxes, as the evaluator allows. A unit test checks the ranks against the evaluator's own matching code on random images with duplicated objects, repeated pairs and relabelled ground truth, for $k \in \{1, 5, 20, 50, 100\}$, and fails on each of three deliberate faults. On the real run the re-scored metrics reproduce the official ones (Tab. 17). Three relations, all in one image with 35 objects, belong to pairs the model never scored; they keep their exact ranks (they can only be matched through duplicates) and are dropped from the probability-based analyses.

Table 17. Official evaluator against the offline replay. The official TDE row is this run's evaluation; the official baseline row is the archived evaluation of the same checkpoint.

Model	R@50	mR@50	ng-R@50	ng-mR@50	mR@100
base, official	0.6612	0.1459	0.8248	0.3260	0.1584
base, re-scored	0.6612	0.1459	0.8248	0.3260	0.1584
TDE, official	0.4588	0.2476	0.5687	0.2981	0.2870
TDE, re-scored	0.4588	0.2476	0.5687	0.2981	0.2870
LA_1, re-scored	0.6437	0.2343	0.8191	0.4173	0.2509
$\mathrm{LA}_{.5}$, re-scored	0.6587	0.1845	0.8255	0.3712	0.1990

Micro-averaged mean recall. The official metric averages each predicate's recall over the images containing it, then over predicates, counting a predicate absent from the test set as zero. Pooling relations instead gives $mR = \sum_i w(y_i)\, h_i(y_i)$ with $w(y) = 1/(K n_y)$, $h_i(y) = \mathbb{1}\{r_i(y) < k\}$ and n_y the number of relations labelled y. With $H_i(y) = N\, w(y)\, [h_i^{(1)}(y) - h_i^{(0)}(y)]$ the estimator of Sec. 3 applies unchanged: Theorems 1 and 3 and the influence function use only the antisymmetric pair structure, never properness, and because transport preserves each cell's label counts it preserves every n_y. We restrict the pooling to identified relations, so that $\widehat{\Delta T}$ is exactly the change in micro-averaged mean recall over the relations the split covers. It tracks the official change ($+0.0972$ against $+0.1017$ at $k = 50$). Restricting w to one tier, or one predicate, splits that tier's or predicate's share; the parts add up across tiers and predicates exactly.

Reading the case-level part of a thresholded metric. Every change that adds a constant to all log-odds within a cell — logit adjustment at any τ, or dropping the frequency branch — leaves the within-cell AUC where it was (Tab. 20; the residue of 10^{-4} is float32 rounding of the stored probabilities) and still shows a significant case-level part of mean recall (Tab. 18): $+0.0064$ and $+0.0139$ for $\mathrm{LA}_{.5}$ and LA_1, $+0.0053$ for removing the frequency branch. The reason is the threshold. A relation is a hit when its own label wins a comparison, and moving the boundary of that comparison flips the relations closest to it; those are the relations the model already scored as most like the newly favoured label, and they carry it more often than the others in their cell. The case-level part of a hit rate thus contains existing discrimination spent at a new operating point. We read it against LA_1, whose group-level change matches TDE's, and use the within-cell AUC as the direct test of new discrimination. Without the graph constraint, where every (pair, predicate) competes for the top k directly, TDE's case-level part against LA_1 is $+0.0072$, interval $[-0.0001, +0.0145]$.

The AUC. For cell c and labels $y \neq z$ present in it, $\mathrm{AUC}_c(y, z)$ is the probability that a relation labelled y has larger $\log q(y) - \log q(z)$ than one labelled z, ties counting one half; the reported AUC averages these with weight $n_c(y) n_c(z)$, the number of comparisons, which is the $p(y)p(z)$ weighting of the pairwise form of ΔR_c. A temperature divides every log-odds by the same constant, and any change of the predicted distribution that is constant within a cell adds the same constant to every log-odds in it; neither changes any within-cell order. The statistic is computed exactly with ties in $O(M \log M)$ for M compared relations; bootstrap replicates re-weight images, so each costs one pass.

The proper-score split in every configuration. The proper-score analyses use the 183,639 relations whose pair was scored. Top-1 predicate accuracy is 0.6981 for the baseline, 0.5577 for TDE and 0.6820 for LA_1; TDE also raises zero-shot recall to $zR@50 = 0.1432$. Table 21 lists the proper-score split of Sec. 5 in every configuration we ran; the matched quadratic case-level part of TDE against the baseline has a cyclic randomization $p = .555$. Raw outputs give a significant case-level *loss*; Sec. 4 showed that a calibration difference alone manufactures one. Grouping by subject alone reverses the sign of the group-level part (Proposition 3). Matched, the quadratic and log scores disagree on TDE's case-level part against the baseline

Table 18. Mean-recall split for every protocol and k. LA_τ: logit-adjusted baseline; ctx, vis+ctx: the baseline's context branch alone and without the frequency branch.

Comparison	Protocol	ΔmR	group	case	case 95% CI
TDE vs base	mR@20	+.0598	+.0462	+.0135	[+.0101, +.0170]
	mR@50	+.0972	+.0841	+.0130	[+.0093, +.0168]
	mR@100	+.1280	+.1099	+.0182	[+.0144, +.0220]
	ng-mR@20	−.0037	−.0134	+.0096	[+.0058, +.0135]
	ng-mR@50	−.0337	−.0464	+.0127	[+.0080, +.0173]
	ng-mR@100	−.0489	−.0724	+.0236	[+.0175, +.0296]
LA_1 vs base	mR@20	+.0831	+.0709	+.0122	[+.0086, +.0158]
	mR@50	+.0941	+.0802	+.0139	[+.0101, +.0176]
	mR@100	+.0981	+.0834	+.0147	[+.0108, +.0187]
	ng-mR@20	+.0754	+.0671	+.0084	[+.0041, +.0126]
	ng-mR@50	+.0935	+.0880	+.0055	[−.0010, +.0120]
	ng-mR@100	+.1061	+.0973	+.0088	[+.0006, +.0170]
$\mathrm{LA}_{.5}$ vs base	mR@20	+.0365	+.0311	+.0055	[+.0033, +.0077]
	mR@50	+.0419	+.0356	+.0064	[+.0040, +.0088]
	mR@100	+.0435	+.0367	+.0068	[+.0044, +.0092]
	ng-mR@20	+.0376	+.0319	+.0057	[+.0022, +.0092]
	ng-mR@50	+.0448	+.0438	+.0010	[−.0026, +.0047]
	ng-mR@100	+.0514	+.0485	+.0029	[−.0033, +.0091]
TDE vs LA_1	mR@20	−.0234	−.0247	+.0013	[−.0024, +.0050]
	mR@50	+.0031	+.0039	−.0008	[−.0050, +.0033]
	mR@100	+.0300	+.0265	+.0034	[−.0010, +.0078]
	ng-mR@20	−.0792	−.0804	+.0013	[−.0035, +.0060]
	ng-mR@50	−.1272	−.1344	+.0072	[−.0001, +.0145]
	ng-mR@100	−.1549	−.1697	+.0148	[+.0064, +.0232]
ctx vs base	mR@20	+.0136	+.0078	+.0058	[+.0040, +.0076]
	mR@50	+.0155	+.0086	+.0068	[+.0047, +.0089]
	mR@100	+.0149	+.0084	+.0065	[+.0042, +.0087]
	ng-mR@20	−.0014	−.0072	+.0058	[+.0030, +.0086]
	ng-mR@50	−.0042	−.0136	+.0094	[+.0058, +.0129]
	ng-mR@100	−.0040	−.0170	+.0130	[+.0089, +.0172]
vis+ctx vs base	mR@20	+.0012	−.0027	+.0040	[+.0026, +.0053]
	mR@50	−.0009	−.0062	+.0053	[+.0036, +.0069]
	mR@100	−.0038	−.0083	+.0045	[+.0029, +.0060]
	ng-mR@20	−.0060	−.0124	+.0065	[+.0040, +.0089]
	ng-mR@50	−.0149	−.0235	+.0086	[+.0056, +.0117]
	ng-mR@100	−.0197	−.0327	+.0129	[+.0091, +.0167]
TDE vs ctx	mR@20	+.0462	+.0384	+.0077	[+.0044, +.0111]
	mR@50	+.0817	+.0755	+.0062	[+.0025, +.0100]
	mR@100	+.1132	+.1015	+.0117	[+.0080, +.0155]
	ng-mR@20	−.0023	−.0061	+.0038	[+.0005, +.0072]
	ng-mR@50	−.0295	−.0328	+.0033	[−.0008, +.0074]
	ng-mR@100	−.0449	−.0554	+.0105	[+.0051, +.0160]

and against LA_1; LA_1, which moves no within-cell ranking, has a positive case-level part under both, so the proper-score split too responds to the operating point.

O.1. TDE along its branches

In PredCls the baseline's logits are $\mathrm{vis} + \mathrm{ctx} + \mathrm{frq}$ and TDE's are $\mathrm{ctx} - \mathrm{ctx}_{\mathrm{avg}}$, so the change runs through two intermediate models built from the same branches: the baseline without the frequency prior ($\mathrm{vis} + \mathrm{ctx}$), and the context term alone. The split is linear in the gain vector, the AUC is a difference of per-model values, and in the matched proper-score split each model keeps one held-out temperature throughout; so along this path every part adds up to TDE's exactly, which the script asserts (Tab. 22). The order of the three steps is a choice, and the steps are attributions along it, not order-free effects.

Two observations make the mechanism plain. Dropping the frequency prior, which is constant within every cell, cannot move the within-cell AUC and does not; it moves mean recall by -0.0009, but shifts the matched quadratic case-level part

Table 19. The $mR@50$ split by tier (training instances: head $>$ 10,000, tail $<$ 500).

Comparison	Tier	$\Delta mR@50$	group	case	case 95% CI
TDE vs base	head	$-.0255$	$-.0241$	$-.0014$	$[-.0019, -.0009]$
	body	$+.1224$	$+.1082$	$+.0142$	$[+.0105, +.0180]$
	tail	$+.0003$	$+.0001$	$+.0002$	$[-.0001, +.0005]$
LA_1 vs base	head	$+.0009$	$+.0006$	$+.0003$	$[-.0000, +.0006]$
	body	$+.0932$	$+.0796$	$+.0136$	$[+.0099, +.0173]$
	tail	$+.0000$	$+.0000$	$+.0000$	$[+.0000, +.0000]$
$LA_{.5}$ vs base	head	$+.0013$	$+.0011$	$+.0002$	$[-.0001, +.0004]$
	body	$+.0407$	$+.0345$	$+.0062$	$[+.0039, +.0085]$
	tail	$+.0000$	$+.0000$	$+.0000$	$[+.0000, +.0000]$
TDE vs LA_1	head	$-.0264$	$-.0248$	$-.0017$	$[-.0021, -.0012]$
	body	$+.0292$	$+.0286$	$+.0006$	$[-.0035, +.0048]$
	tail	$+.0003$	$+.0001$	$+.0002$	$[-.0001, +.0005]$

Table 20. Within-cell AUC (comparison-weighted), image bootstrap with 200 replicates.

Comparison	AUC new	AUC old	ΔAUC	95% CI
TDE vs base	0.5731	0.5705	$+.0026$	$[-.0028, +.0086]$
LA_1 vs base	0.5705	0.5705	$+.0001$	$[-.0000, +.0001]$
$LA_{.5}$ vs base	0.5705	0.5705	$+.0000$	$[-.0000, +.0001]$
TDE vs LA_1	0.5731	0.5705	$+.0026$	$[-.0029, +.0086]$
TDE vs ctx	0.5731	0.5715	$+.0016$	$[+.0008, +.0025]$
vis+ctx vs base	0.5705	0.5705	$+.0001$	$[-.0000, +.0001]$
ctx vs base	0.5715	0.5705	$+.0010$	$[-.0042, +.0068]$
vis vs base	0.5578	0.5705	$-.0127$	$[-.0163, -.0090]$
IETrans vs base	0.5612	0.5705	$-.0093$	$[-.0181, +.0004]$
IETrans vs LA_1	0.5612	0.5705	$-.0093$	$[-.0182, +.0002]$
IETrans vs TDE	0.5612	0.5731	$-.0119$	$[-.0201, -.0022]$
IETrans vs IETrans w/o prior	0.5612	0.5612	$+.0000$	$[+.0000, +.0000]$

Table 21. Proper-score split of TDE and of the logit-adjusted baseline (LA), every configuration. Matched rows fit each model's temperature on a random half of the images and audit the other half.

Comparison	Outputs	Score	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$	$\widehat{\Delta R}$ 95% CI
TDE vs base	raw	quad.	-0.11651	-0.10296	-0.01355	$[-0.01504, -0.01206]$
TDE vs base	raw	log	$+0.03277$	$+0.15897$	-0.12619	$[-0.13093, -0.12145]$
TDE vs base	raw, subject only	quad.	-0.11536	$+0.10175$	-0.21711	$[-0.22124, -0.21298]$
TDE vs base	raw, audit half	quad.	-0.11548	-0.10376	-0.01172	$[-0.01374, -0.00970]$
TDE vs base	matched	quad.	-0.18264	-0.18258	-0.00006	$[-0.00120, +0.00109]$
TDE vs base	matched	log	-0.54760	-0.59157	$+0.04396$	$[+0.04079, +0.04714]$
LA_1 vs base	matched	quad.	-0.02503	-0.02895	$+0.00392$	$[+0.00334, +0.00450]$
LA_1 vs base	matched	log	-0.05095	-0.07947	$+0.02852$	$[+0.02719, +0.02985]$
TDE vs LA_1	matched	quad.	-0.15761	-0.15363	-0.00398	$[-0.00510, -0.00285]$
TDE vs LA_1	matched	log	-0.49665	-0.51209	$+0.01544$	$[+0.01282, +0.01806]$
$LA_{.5}$ vs base	matched	quad.	-0.00358	-0.00573	$+0.00215$	$[+0.00188, +0.00242]$
$LA_{.5}$ vs base	matched	log	-0.00209	-0.01398	$+0.01189$	$[+0.01133, +0.01244]$
IETrans vs base	matched	quad.	-0.17695	-0.16071	-0.01623	$[-0.01767, -0.01480]$
IETrans vs base	matched	log	-0.48127	-0.42692	-0.05435	$[-0.05786, -0.05085]$
IETrans vs LA_1	matched	quad.	-0.15191	-0.13176	-0.02015	$[-0.02174, -0.01857]$
IETrans vs LA_1	matched	log	-0.43032	-0.34745	-0.08287	$[-0.08747, -0.07828]$
IETrans vs TDE	matched	quad.	$+0.00569$	$+0.02187$	-0.01618	$[-0.01747, -0.01488]$
IETrans vs TDE	matched	log	$+0.06633$	$+0.16465$	-0.09832	$[-0.10388, -0.09275]$

by $+0.00915$, another instance of the operating-point effect. And the averaged-context term that TDE subtracts is almost a constant: splitting each logit term's energy, after removing each row's mean, into the vector shared by every relation, what the subject–object cell adds, and what varies within cells (Tab. 23), 99.96% of $\mathrm{ctx}_{\mathrm{avg}}$ is the shared vector, with a non-shared remainder of 0.021 logits against a size of 1.10. The shared vector correlates $+0.75$ with the log training prior. Subtracting it

Table 22. TDE's change from the baseline, step by step (graph-constrained $mR@50$ above; no-graph-constraint $mR@50$ and the matched quadratic-score case-level part below). Each column sums over the steps to TDE's row in Tabs. 18 and 21.

Step	$\Delta mR@50$	group	case	case 95% CI	ΔAUC	95% CI
drop the frequency prior	$-.0009$	$-.0062$	$+.0053$	$[+.0036, +.0069]$	$+.0001$	$[-.0000, +.0001]$
drop the visual term	$+.0164$	$+.0148$	$+.0016$	$[-.0006, +.0037]$	$+.0010$	$[-.0044, +.0068]$
subtract the averaged context	$+.0817$	$+.0755$	$+.0062$	$[+.0025, +.0100]$	$+.0016$	$[+.0008, +.0025]$
No graph constraint; last two columns: matched quadratic-score case-level part						
drop the frequency prior	$-.0149$	$-.0235$	$+.0086$	$[+.0056, +.0117]$	$+0.00915$	$[+0.00852, +0.00979]$
drop the visual term	$+.0106$	$+.0099$	$+.0007$	$[-.0026, +.0040]$	-0.00178	$[-0.00283, -0.00074]$
subtract the averaged context	$-.0295$	$-.0328$	$+.0033$	$[-.0008, +.0074]$	-0.00743	$[-0.00845, -0.00641]$

is therefore, to first order, a logit adjustment of the context branch by a learned prior; its small remainder is what moves the within-cell AUC by $+0.0016$.

Table 23. Energy of each logit term across the 183,639 scored relations: shared by all relations (global), set by the subject–object cell (between), varying within cells (within); root-mean-square size of the term and of its non-global part; correlation of the global vector with the log training prior $\log \pi$. The frequency term's within-cell share is zero up to float32 rounding.

Logit term	rms	global	between	within	non-global rms	corr. with $\log \pi$
frq	0.98	0.2976	0.7023	0.0001	0.820	$+0.779$
vis	2.07	0.4875	0.4126	0.0999	1.485	$+0.913$
ctx(post)	2.34	0.3891	0.5614	0.0495	1.830	$+0.839$
ctx(avg)	1.10	0.9996	0.0001	0.0003	0.021	$+0.754$
TDE	1.94	0.1045	0.8229	0.0726	1.831	$+0.631$
baseline	5.07	0.4441	0.5178	0.0381	3.778	$+0.878$

O.2. The IETrans checkpoint

What was run. The authors release one Neural-Motifs PredCls checkpoint for VG150 (their model zoo, Google Drive), saved at iteration 30,000. It is the IETrans+Rwt variant: the evaluation command the authors give for it (`cmds/50/motif/predcls/lt/combine/val.sh`) reads the output of their `combine/rwt` training, internal and external transfer followed by a reweighted loss, and its official numbers ($R@50 = 0.4858$, $mR@50 = 0.3576$ on their test set) are those of their IETrans+Rwt row. We call it IETrans+Rwt throughout; its effect cannot be attributed to relabelling alone. It holds no frequency-bias weights: the model was trained without one. Their evaluation command builds the bias at test time from the training split's log predicate frequencies per object-class pair, so the released model's logit is a learned context term plus a fixed per-cell log-prior; we ran exactly that command, with the dump hook of Appendix O recording both terms and their sum (equal to the returned logits with largest difference 0). Their annotation file matches the standard VG150 file array for array, except that 5,000 training images were moved to a validation split; the test split is identical.

Test relations. IETrans's data loader removes exact duplicate relations — the same subject object, object object and predicate annotated twice — from the test set as well as the training set, which leaves 152,226 test relations where the standard protocol keeps 183,642. A duplicate shares its pair and its label with the relation that was kept, so the evaluator matches it at the same ranks and the model scores it with the same probabilities. We therefore mapped every standard relation to the IETrans row with the same image, boxes, classes and predicate: the map is total, uses every IETrans row, and no key is ambiguous. Table 24 reports IETrans on both test sets; the difference is small, and every comparison in the paper uses the standard relations.

Table 24. The released IETrans checkpoint under the official recall rules, on its own test relations and on the standard ones.

Test relations	R@50	mR@20	mR@50	mR@100	ng-mR@50
IETrans's (152,226)	0.4858	0.2884	0.3576	0.3905	0.5278
standard (183,642)	0.4891	0.2887	0.3587	0.3906	0.5291

Table 25. Mean-recall split of the IETrans checkpoint against the MOTIFS baseline (with tiers), the logit-adjusted baseline, TDE, and itself without the test-time prior. The proper-score rows are in Tab. 21, the AUC rows in Tab. 20.

Comparison	Protocol / tier	ΔmR	group	case	case 95% CI
IETrans vs base	mR@50	+.2204	+.2143	+.0060	[−.0031, +.0151]
	ng-mR@50	+.2069	+.2087	−.0018	[−.0134, +.0099]
	mR@50 head	−.0169	−.0132	−.0038	[−.0042, −.0033]
	mR@50 body	+.1484	+.1429	+.0055	[+.0020, +.0090]
	mR@50 tail	+.0889	+.0846	+.0043	[−.0041, +.0127]
IETrans vs LA_1	mR@50	+.1263	+.1341	−.0079	[−.0173, +.0015]
	ng-mR@50	+.1134	+.1206	−.0072	[−.0185, +.0040]
IETrans vs TDE	mR@50	+.1232	+.1302	−.0070	[−.0164, +.0023]
	ng-mR@50	+.2406	+.2550	−.0144	[−.0261, −.0028]
IETrans vs IETrans w/o prior	mR@50	−.0133	−.0046	−.0088	[−.0171, −.0004]
	ng-mR@50	−.0145	−.0026	−.0119	[−.0205, −.0032]

Reading it. Adding the test-time prior to IETrans's context term is a change constant within every cell: it leaves the within-cell AUC exactly where it was, costs 0.0133 of mean recall, and its case-level part, −0.0088, is again an operating-point effect. Against the baseline, the mean-recall gain is 97% group-level and its case-level part is not distinguishable from zero, while both matched proper scores give a case-level loss (Tab. 21). The comparison is between models of the same family trained separately: IETrans released no plain baseline, and the MOTIFS-SUM baseline of Sec. 5 differs from IETrans's Motifs predictor in its fusion of the visual term.

P. The real-pixel study across seeds

Every head of Sec. 6 was retrained with five seeds, which set the initialisation and the batch order; the training subsample of 100,000 relations, the CLIP features and the test set are the same throughout, and seed 0 reproduces the reported predictions exactly (largest absolute difference 0). A fourth head, MLP-VISGEO-S, receives the class embeddings, the twelve geometry features and the CLIP crop features together, so that what pixels add *to* the geometric proxy is measured directly. Each row of Tab. 26 uses the paper's protocol: temperatures fitted on the same held-out half of the images, the split on the other half, and the within-cell AUC with an image bootstrap of 200 replicates.

Table 26. Matched quadratic-score split and within-cell AUC for five training seeds.

Comparison	Seed	acc. new	acc. old	group	case	case 95% CI	ΔAUC [95% CI]
CLIP vs geometry	0	0.6156	0.6467	−0.04705	+0.00464	[+0.00311, +0.00618]	+.0085 [−.0062, +.0214]
	1	0.6165	0.6474	−0.04861	+0.00474	[+0.00321, +0.00626]	+.0092 [−.0075, +.0236]
	2	0.6174	0.6462	−0.04629	+0.00450	[+0.00298, +0.00603]	+.0093 [−.0098, +.0238]
	3	0.6148	0.6470	−0.04857	+0.00413	[+0.00258, +0.00568]	+.0156 [+.0031, +.0298]
	4	0.6162	0.6468	−0.04857	+0.00441	[+0.00288, +0.00594]	+.0048 [−.0098, +.0185]
CLIP+geometry vs geometry	0	0.6215	0.6467	−0.04466	+0.00949	[+0.00797, +0.01101]	+.0125 [−.0013, +.0278]
	1	0.6220	0.6474	−0.04525	+0.00894	[+0.00746, +0.01041]	+.0179 [+.0023, +.0338]
	2	0.6225	0.6462	−0.04344	+0.00870	[+0.00721, +0.01019]	+.0099 [−.0060, +.0231]
	3	0.6206	0.6470	−0.04498	+0.00807	[+0.00658, +0.00955]	+.0086 [−.0043, +.0220]
	4	0.6226	0.6468	−0.04438	+0.00874	[+0.00726, +0.01023]	+.0047 [−.0088, +.0192]
CLIP+geometry vs CLIP	0	0.6215	0.6156	+0.00240	+0.00485	[+0.00364, +0.00605]	+.0040 [−.0045, +.0142]
	1	0.6220	0.6165	+0.00336	+0.00420	[+0.00303, +0.00538]	+.0088 [+.0001, +.0173]
	2	0.6225	0.6174	+0.00285	+0.00419	[+0.00298, +0.00540]	+.0007 [−.0072, +.0090]
	3	0.6206	0.6148	+0.00359	+0.00393	[+0.00273, +0.00514]	−.0070 [−.0178, +.0018]
	4	0.6226	0.6162	+0.00419	+0.00433	[+0.00313, +0.00553]	−.0001 [−.0082, +.0080]

The case-level advantage of pixels over geometry is stable across seeds, in contrast with the roughly fourfold seed variation of the CIFAR-100-LT effect (Appendix N.5); so is the accuracy deficit. The AUC difference has the same sign in every seed and is small against its bootstrap interval, which is why the paper reports the covariance and not better recognition. The combined head loses at group level almost as much as the CLIP head does against geometry, and gains about twice its case-level part.

Table 27. **The matched split over 20 calibration halves**: case-level part, its mean, SD and range over halves, and the number of halves whose image-clustered interval excludes zero. "One T": a single temperature for TDE and the baseline.

Comparison	Score	mean	sd	min	max	CI $\not\ni 0$
TDE vs base (T+bias)	quadratic	$+0.00397$	0.00040	$+0.00341$	$+0.00482$	20/20
	log	$+0.02462$	0.00089	$+0.02367$	$+0.02629$	20/20
LA_1 vs base (T+bias)	quadratic	-0.00000	0.00000	-0.00000	$+0.00000$	–
	log	-0.00000	0.00001	-0.00002	$+0.00001$	11/20
IETrans vs base (T+bias)	quadratic	-0.01709	0.00057	-0.01834	-0.01619	20/20
	log	-0.06514	0.00160	-0.06825	-0.06279	20/20
TDE vs base (T)	quadratic	-0.00057	0.00055	-0.00131	$+0.00059$	3/20
	log	$+0.04315$	0.00159	$+0.04142$	$+0.04761$	20/20
LA_1 vs base (T)	quadratic	$+0.00375$	0.00027	$+0.00325$	$+0.00429$	20/20
	log	$+0.02618$	0.00145	$+0.02390$	$+0.02793$	20/20
IETrans vs base (T)	quadratic	-0.01662	0.00073	-0.01806	-0.01538	20/20
	log	-0.05300	0.00203	-0.05629	-0.04923	20/20
TDE vs base, one T	quadratic	-0.01922	0.00071	-0.02035	-0.01738	20/20
	log	-0.09064	0.00183	-0.09440	-0.08727	20/20

Q. How robust the scene-graph audit is

Three choices in Sec. 5 could carry its conclusions: the random half of the test images on which each model's temperature is fitted, the 50 predicates themselves, several of which are near-synonyms, and the label mix of the test set. This section varies each. Scripts: `run_sgg_split_robustness.py`, `run_predicate_merge.py`, `run_label_shift.py`; tables are generated from their results files.

Q.1. Calibration halves and a shared temperature

The confidence-matched rows of Sec. 5 fit each model's temperature on a random half of the test images and split the other half. The released checkpoints come with no validation-set outputs, and we did not rerun them on the validation split; the calibration half shares no image with the audited half, so no relation is used both to fit a temperature and to be scored. To see how much the result depends on which half, Tab. 27 repeats the split for 20 independent halves. The fitted temperatures move by at most 0.06 across halves. Under the primary protocol (a temperature and a per-class bias per model) TDE's quadratic case-level part is positive in all twenty halves, from $+0.00341$ to $+0.00482$, its interval excluding zero every time; adding the spread over halves to the sampling error in quadrature gives a half-width of 0.0012 around the mean $+0.00397$. The control's part is zero to within 10^{-5} in every half. With a temperature alone TDE's part ranges from -0.00131 to $+0.00059$; its interval excludes zero in three halves, each time on the negative side. Under the log score it is positive in all twenty ($+0.04142$ to $+0.04761$). The control's and IETrans's parts keep their sign, and exclude zero, in every half under both scores.

TDE and the baseline share weights, so one may prefer a single temperature for both, fitted on the pooled calibration half. That temperature (1.47) lies between the two models' own optima (0.92 and 2.50), so neither model is confidence-matched, and TDE's case-level part turns negative under both scores in every half (quadratic -0.02035 to -0.01738, log -0.09440 to -0.08727). The sign of TDE's part is therefore set by the protocol, and the primary one was fixed by the criterion of Sec. 3 before these runs.

Q.2. Merging near-synonymous predicates

A case-level part near zero could reflect predicates that no model can tell apart from the image: on, sitting on and standing on; wearing and wears; has, with and of. We apply two merges declared before looking at their results. *Synonym* merges eleven placement verbs into on, three in forms, the two wear forms, and six possessive forms into has, leaving every other predicate itself (32 classes). *Coarse* collapses the 50 predicates into three families: geometric, possessive and semantic. A model's probability for a merged class is the sum over its members; for recall, a relation counts as a hit for a merged class when any member would (the smallest of the members' first-hit ranks). Temperatures are fitted on the unmerged distributions on the audit's calibration half. We also restrict to the $166{,}940$ of $183{,}639$ relations whose subject–object box pair carries a single annotated predicate, so that no second annotated predicate competes with the label.

Table 28 reports the result. No conclusion of Sec. 5 depends on the predicate vocabulary. TDE's matched case-level part

Table 28. **The audit after merging predicates** (number of classes in brackets). Matched quadratic case-level part on the audit half, on all relations and on single-annotation box pairs; within-cell AUC change; mean recall@50 split (graph constraint) with its image-clustered case-level interval. Every row is against the MOTIFS baseline.

Merge	vs base	case	95% CI	case, single	ΔAUC [95% CI]	ΔmR	group	case [95% CI]
none (50)	TDE	-0.00006	$[-0.00120, +0.00109]$	$+0.00007$	$+.0026$ $[-.0028, +.0082]$	$+.0972$	$+.0841$	$+.0130$ $[+.0093, +.0168]$
	LA_1	$+0.00392$	$[+0.00334, +0.00450]$	$+0.00384$	$+.0001$ $[-.0000, +.0001]$	$+.0941$	$+.0802$	$+.0139$ $[+.0101, +.0176]$
	IETrans	-0.01623	$[-0.01767, -0.01480]$	-0.01625	$-.0093$ $[-.0177, +.0013]$	$+.2204$	$+.2143$	$+.0060$ $[-.0031, +.0151]$
synonym (32)	TDE	$+0.00059$	$[-0.00050, +0.00168]$	$+0.00104$	$-.0000$ $[-.0072, +.0083]$	$+.0865$	$+.0705$	$+.0160$ $[+.0106, +.0214]$
	LA_1	$+0.00419$	$[+0.00369, +0.00469]$	$+0.00423$	$-.0012$ $[-.0024, -.0003]$	$+.0921$	$+.0758$	$+.0164$ $[+.0121, +.0206]$
	IETrans	-0.01540	$[-0.01675, -0.01405]$	-0.01529	$-.0127$ $[-.0236, +.0019]$	$+.1850$	$+.1788$	$+.0062$ $[-.0038, +.0162]$
coarse (3)	TDE	-0.00015	$[-0.00095, +0.00065]$	$+0.00006$	$+.0045$ $[-.0045, +.0136]$	$+.0324$	$+.0231$	$+.0093$ $[+.0037, +.0149]$
	LA_1	$+0.00175$	$[+0.00137, +0.00213]$	$+0.00172$	$+.0008$ $[+.0001, +.0012]$	$+.1236$	$+.1159$	$+.0077$ $[+.0028, +.0127]$
	IETrans	-0.00761	$[-0.00868, -0.00653]$	-0.00734	$-.0027$ $[-.0147, +.0144]$	$+.1359$	$+.1422$	$-.0063$ $[-.0108, -.0017]$

stays null under both merges ($+0.00059$, interval $[-0.00050, +0.00168]$; -0.00015, $[-0.00095, +0.00065]$) and on single-annotation pairs, and the interval of its AUC change covers zero under both. IETrans's stays negative and excludes zero under every merge (-0.01540 synonym, -0.00761 coarse); under the coarse merge its mean-recall case-level part also turns significantly negative (-0.0063). The logit-adjusted control keeps a positive matched case-level part throughout. Summing probabilities over a merged class can reorder the merged classes within a cell, so under a merge the control's AUC is no longer fixed by construction; it moves by at most 0.0012.

Could the benchmark leave little room for any case-level gain? One kind of ceiling is measurable: two relations with the same image and the same subject and object boxes but different predicates receive the same prediction from every model that sees the image and the boxes, so every comparison between them is a tie. They carry 0.07% of the AUC's comparison weight (0.64% weighted as the split weights cells), which caps the within-cell AUC at 0.9996 (`sgg_auc_ceiling.py`). MOTIFS's 0.5705 is far below that; how much of the gap is annotator style rather than missing recognition, the data cannot say.

Q.3. A label-shifted test

Section 3 writes a cell's case-level part as $\Delta R_c = \sum_{y<z} p(y)p(z)\, D(y,z)$, where $D(y,z)$ depends only on the class-conditional behaviour of the gain. A label shift that leaves $P(X \mid Y, C)$ fixed changes the weights $p(y)p(z)$ and no D, while the group-level part depends on the cell's label frequencies directly. So a case-level part keeps its sign under every such shift if its $D(y,z)$ share one sign, and can change sign otherwise; that much follows from the identity, and the reweighted estimator below inherits it. What the data decide is which comparisons have $D(y,z)$ of mixed sign, and how far the group-level parts move. This is a sensitivity analysis, not an independent test.

Relation i in cell c gets weight $w_i = 1/(L_c\, \hat{p}_c(y_i))$, with L_c the number of predicates present in the cell and $\hat{p}_c$ their frequencies, so that every predicate present in a cell carries equal total weight; the path w_i^{α}, $\alpha \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, moves from the benchmark's mix to that uniform one. The split is recomputed with pairs (i,j) of a cell weighted by $w_i w_j$, so a gain constant within a cell still has a case-level part of exactly zero and unit weights give the estimator of Sec. 3; the within-cell AUC is recomputed with the same comparison weights. Temperatures are those fitted at the benchmark's mix, on the calibration halves of Secs. 5 and 6, each study on its own temperature grid; they are not refitted under the shift. Intervals are a delete-a-group jackknife over images (100 groups). An image bootstrap biases the case-level part towards zero here, because a resampled image's same-label pairs have kernel zero and receive weight $\mathbb{E}[w^2] = 2$. At the uniform end the weights are very unequal: the Kish effective sample size of the audit half falls from $92{,}027$ to $7{,}296$.

Table 29 gives the result for the eight comparisons of the two audits. The group-level part changes sign in seven of them; in the eighth, the class-pair model against FREQ, it shrinks from -0.01432 to -0.00241. With the group-level parts, the totals reverse too: the CLIP model, last on the benchmark by every aggregate, beats geometry by $+0.02981$ under uniform labels. Of the six case-level parts whose interval excludes zero on the benchmark, five keep their sign and grow in size: geometry over the class-pair model ($+0.00891 \to +0.01157$), CLIP over geometry ($+0.00464 \to +0.00615$), IETrans+Rwt over the baseline ($-0.01623 \to -0.01830$). The exception is TDE against the logit-adjusted control, whose case-level part turns from -0.00398 to $+0.00723$; and TDE's null against the baseline becomes $+0.01430$, interval $[+0.01116, +0.01744]$. Along the path (Tab. 30) TDE's part rises monotonically and already excludes zero at $\alpha = \frac{1}{4}$ ($+0.00238$, $[+0.00092, +0.00385]$): its null holds at the benchmark's exact mix and nowhere near it. Both exceptions mean that TDE's $D(y,z)$ have mixed signs:

Table 29. **The split under a label shift.** Matched quadratic split on the benchmark's labels and with every predicate of a cell given equal weight. Upper block: the scene-graph audit (Sec. 5); middle block: the real-pixel study (Sec. 6); lower block: within-cell AUC change under the same two weightings (image bootstrap). Split intervals: delete-a-group jackknife over images.

	benchmark labels			uniform within cells		
Comparison	group	case	case 95% CI	group	case	case 95% CI
TDE vs base	-0.18258	-0.00006	$[-0.00125, +0.00114]$	$+0.08762$	$+0.01430$	$[+0.01116, +0.01744]$
LA_1 vs base	-0.02895	$+0.00392$	$[+0.00329, +0.00455]$	$+0.00340$	$+0.00707$	$[+0.00613, +0.00801]$
IETrans vs base	-0.16071	-0.01623	$[-0.01759, -0.01488]$	$+0.07342$	-0.01830	$[-0.02086, -0.01574]$
TDE vs LA_1	-0.15363	-0.00398	$[-0.00518, -0.00277]$	$+0.08422$	$+0.00723$	$[+0.00436, +0.01011]$
class vs FREQ	-0.01432	-0.00000	$[-0.00000, +0.00000]$	-0.00241	$+0.00000$	$[-0.00000, +0.00000]$
geometry vs class	-0.00507	$+0.00891$	$[+0.00812, +0.00971]$	$+0.00118$	$+0.01157$	$[+0.01020, +0.01294]$
CLIP vs class	-0.05213	$+0.01356$	$[+0.01190, +0.01522]$	$+0.02484$	$+0.01772$	$[+0.01540, +0.02004]$
CLIP vs geometry	-0.04705	$+0.00464$	$[+0.00300, +0.00628]$	$+0.02365$	$+0.00615$	$[+0.00388, +0.00842]$
	ΔAUC, benchmark labels			ΔAUC, uniform within cells		
TDE vs base	$+.0026$ $[-.0028, +.0082]$			$-.0183$ $[-.0684, +.0082]$		
LA_1 vs base	$+.0001$ $[-.0000, +.0001]$			$+.0020$ $[-.0015, +.0054]$		
IETrans vs base	$-.0093$ $[-.0177, +.0013]$			$+.0082$ $[-.0471, +.0607]$		

Table 30. **Case-level part along the path of label mixes** w^α ($\alpha = 0$: the benchmark; $\alpha = 1$: uniform within cells), matched quadratic score, jackknife intervals; last row: Kish effective sample size.

α	0	.25	.5	.75	1
TDE vs base	-0.00006	$+0.00238$	$+0.00603$	$+0.01039$	$+0.01430$
	$[-0.00125, +0.00114]$	$[+0.00092, +0.00385]$	$[+0.00420, +0.00787]$	$[+0.00802, +0.01276]$	$[+0.01116, +0.01744]$
LA_1 vs base	$+0.00392$	$+0.00511$	$+0.00624$	$+0.00699$	$+0.00707$
	$[+0.00329, +0.00455]$	$[+0.00439, +0.00583]$	$[+0.00544, +0.00704]$	$[+0.00612, +0.00787]$	$[+0.00613, +0.00801]$
IETrans+Rwt vs base	-0.01623	-0.01891	-0.02073	-0.02069	-0.01830
	$[-0.01759, -0.01488]$	$[-0.02056, -0.01726]$	$[-0.02272, -0.01874]$	$[-0.02299, -0.01839]$	$[-0.02086, -0.01574]$
TDE vs LA_1	-0.00398	-0.00273	-0.00021	$+0.00339$	$+0.00723$
	$[-0.00518, -0.00277]$	$[-0.00417, -0.00128]$	$[-0.00198, +0.00157]$	$[+0.00116, +0.00563]$	$[+0.00436, +0.01011]$
Kish n	92,027	84,921	59,464	25,778	7,296

it gains within-cell discrimination between some pairs of predicates and loses it between others, consistent with the tier pattern of Appendix Q.4. The threshold-free AUC does not separate TDE from the baseline at either end: $+0.0026$ on the benchmark, -0.0183, interval $[-0.0684, +0.0082]$, under uniform labels.

Q.4. How the case-level verdicts depend on the measure

AUC weighting. The within-cell AUC of Sec. 3 averages the AUCs of each cell's label pairs with weight $n_c(y)n_c(z)$, the number of comparisons. Across cells this weights a cell by about n_c^2: the largest cell carries 27% of the weight, against 2.5% of the relations. Weighting by $n_c(y)n_c(z)/n_c$ instead combines cells as the case-level part does. Table 31 gives both, on all identified relations and on the audit half on which the matched split is computed. TDE's change covers zero under all four. The logit-adjusted control's moves by float32 rounding only. IETrans+Rwt's loss excludes zero under the split's weighting (-0.0235, $[-0.0275, -0.0182]$) though not by comparison. CLIP against geometry is the case the weighting decides: by comparison it favours CLIP on the audit half ($+0.0214$, $[+0.0001, +0.0435]$), by relation it favours geometry on all relations (-0.0067, $[-0.0120, -0.0019]$), and the other two estimates cover zero.

Where TDE's discrimination moved. Table 32 restricts the comparison-weighted AUC change to label pairs by predicate tier. 65% of the weight sits on head–head pairs, where TDE's change is $+0.0068$, $[-0.0003, +0.0134]$; on all other pairs it is -0.0051, $[-0.0130, +0.0026]$. Neither excludes zero, so we read it as a pattern consistent with the label-shift result rather than as a finding. IETrans+Rwt's loss is concentrated on head–body (-0.0250) and body–body (-0.0260) pairs, both excluding zero.

Recalibration family. Table 33 repeats the matched split under three families, each fitted on the calibration half and each preserving every model's within-cell ranking: one temperature per model, one temperature shared by the two models, and a

Table 31. **Within-cell AUC change under both weightings**, on all identified relations and on the audit half; image bootstrap, 200 replicates. Lower block: the real-pixel study (seed 0).

Comparison	weighting	all relations	audit half
TDE vs base	comparison	$+.0026$ $[-.0028, +.0086]$	$-.0010$ $[-.0082, +.0062]$
	relation	$-.0015$ $[-.0036, +.0018]$	$-.0016$ $[-.0049, +.0023]$
TDE vs LA_1	comparison	$+.0026$ $[-.0029, +.0086]$	$-.0010$ $[-.0081, +.0064]$
	relation	$-.0018$ $[-.0039, +.0014]$	$-.0016$ $[-.0050, +.0025]$
LA_1 vs base	comparison	$+.0001$ $[-.0000, +.0001]$	$-.0000$ $[-.0002, +.0001]$
	relation	$+.0003$ $[+.0001, +.0006]$	$+.0000$ $[-.0004, +.0004]$
IETrans+Rwt vs base	comparison	$-.0093$ $[-.0181, +.0004]$	$-.0114$ $[-.0272, +.0020]$
	relation	$-.0235$ $[-.0275, -.0182]$	$-.0227$ $[-.0293, -.0152]$
CLIP vs geometry	comparison	$+.0085$ $[-.0062, +.0214]$	$+.0214$ $[+.0001, +.0435]$
	relation	$-.0067$ $[-.0120, -.0019]$	$+.0000$ $[-.0067, +.0068]$
geometry vs class	comparison	$+.0323$ $[+.0210, +.0424]$	$+.0191$ $[+.0037, +.0351]$
	relation	$+.0626$ $[+.0572, +.0644]$	$+.0567$ $[+.0488, +.0592]$
CLIP vs class	comparison	$+.0408$ $[+.0292, +.0519]$	$+.0404$ $[+.0226, +.0576]$
	relation	$+.0559$ $[+.0508, +.0580]$	$+.0567$ $[+.0487, +.0600]$

Table 32. **Comparison-weighted AUC change by predicate-tier pair** (head: more than 10,000 training instances; tail: fewer than 500), with each tier pair's share of the comparison weight; image bootstrap, 200 replicates.

Tier pair	weight	TDE vs base	LA_1 vs base	IETrans+Rwt vs base
head-head	0.649	$+.0068$ $[-.0003, +.0134]$	$+.0000$ $[-.0000, +.0001]$	$-.0015$ $[-.0127, +.0096]$
head-body	0.317	$-.0057$ $[-.0140, +.0022]$	$+.0001$ $[-.0001, +.0003]$	$-.0250$ $[-.0399, -.0117]$
head-tail	0.021	$+.0058$ $[-.0248, +.0355]$	$+.0002$ $[-.0031, +.0029]$	$+.0003$ $[-.0599, +.0614]$
body-body	0.011	$-.0086$ $[-.0184, +.0010]$	$+.0004$ $[-.0002, +.0024]$	$-.0260$ $[-.0405, -.0119]$
body-tail	0.002	$+.0081$ $[-.0217, +.0400]$	$+.0008$ $[-.0050, +.0053]$	$-.0485$ $[-.0961, +.0109]$
tail-tail	0.000	$-.0675$ $[-.1918, +.0494]$	$+.0087$ $[+.0007, +.0297]$	$-.1629$ $[-.3447, +.0122]$
all but head-head	0.351	$-.0051$ $[-.0130, +.0026]$	$+.0001$ $[-.0001, +.0004]$	$-.0237$ $[-.0372, -.0113]$

temperature with a per-class bias per model. The criterion of Sec. 3 picks the third as primary: it absorbs any global shift of the log-prior, so the logit-adjusted control becomes the baseline exactly (case-level part 0.00000), which the first cannot achieve ($+0.00392$); the simulation of Sec. 4 shows the same for a synthetic class-wise offset, which a temperature leaves at $+0.0106$ and the third family removes. Under it TDE's case-level part is $+0.00445$, $[+0.00354, +0.00536]$ (quadratic) and $+0.02543$ (log), while IETrans+Rwt's loss is unchanged (-0.01679, -0.06609). IETrans+Rwt's case-level loss is negative under all six protocol-score combinations; TDE's sign is not. For the CLIP model against geometry (Sec. 6) the three families give $+0.00472$, $+0.00604$ and $+0.00331$, $[+0.00185, +0.00477]$, all excluding zero.

Equivalence margins. A non-detection is bounded as follows (`sgg_equivalence.py`). The image-bootstrap interval gives each AUC contrast a standard error; the minimal detectable change at 80% power and 5% size is 2.80 standard errors; and an equivalence margin declared before the audit, one tenth of the AUC that box geometry buys over the class pair under the same weighting ($+0.0626$ relation, $+0.0323$ comparison), is tested by two one-sided tests: equivalence holds when the 90% interval lies inside the margin. For TDE against the baseline the relation-weighted contrast has standard error 0.0014, minimal detectable change 0.0038, margin ± 0.0063 and 90% interval $[-0.0037, +0.0008]$: equivalent. By comparison the standard error is 0.0029, the minimal detectable change 0.0082, the margin ± 0.0032 and the interval $[-0.0022, +0.0074]$: not equivalent, and not a detection. Against the control the figures are the same to the fourth decimal. IETrans+Rwt's relation-weighted loss, $[-0.0274, -0.0196]$, lies outside its margin on the negative side: a detected loss. TDE's mean-recall case-level part against the control, standard error 0.0021, is equivalent to zero within a tenth of TDE's gain (± 0.0097; 90% interval $[-0.0043, +0.0027]$).

Q.5. Beyond PredCls: SGCls and SGDet

The causal MOTIFS-SUM release includes SGCls and SGDet checkpoints alongside the PredCls one. Under SUM fusion TDE's logit is $\mathrm{ctx}(\mathrm{post}) - \mathrm{ctx}(\mathrm{avg})$ in every protocol, so one TDE pass again yields the baseline ($\mathrm{vis} + \mathrm{ctx}(\mathrm{post}) + \mathrm{frq}$ of the predicted labels) and the logit-adjusted control, and all three share every object prediction: the boxes, labels and scores come from the factual object branch. A pass stores logits for up to 6,320 pairs per image in SGDet, too many to keep, so the evaluator is replayed inside the inference loop (`colab_sgg_inline.py`): for each ground-truth relation and each

Table 33. **The matched split under three recalibration families**, audit half, image-clustered intervals. Every family preserves each model's within-cell ranking.

Comparison	Matching	group	case	case 95% CI
TDE vs base, quadratic	temperature	-0.18258	-0.00006	$[-0.00120, +0.00109]$
	shared temperature	-0.18295	-0.01839	$[-0.02015, -0.01663]$
	temperature + class bias	-0.01396	$+0.00445$	$[+0.00354, +0.00536]$
TDE vs base, log	temperature	-0.59157	$+0.04396$	$[+0.04079, +0.04714]$
	shared temperature	-0.43284	-0.09280	$[-0.09750, -0.08810]$
	temperature + class bias	-0.05861	$+0.02543$	$[+0.02287, +0.02800]$
LA_1 vs base, quadratic	temperature	-0.02895	$+0.00392$	$[+0.00334, +0.00450]$
	shared temperature	-0.02287	-0.00080	$[-0.00134, -0.00025]$
	temperature + class bias	-0.00000	$+0.00000$	$[-0.00000, +0.00000]$
LA_1 vs base, log	temperature	-0.07947	$+0.02852$	$[+0.02719, +0.02985]$
	shared temperature	-0.05057	$+0.00000$	$[-0.00000, +0.00001]$
	temperature + class bias	$+0.00000$	$+0.00000$	$[-0.00000, +0.00000]$
IETrans+Rwt vs base, quadratic	temperature	-0.16071	-0.01623	$[-0.01767, -0.01480]$
	shared temperature	-0.14554	-0.02191	$[-0.02359, -0.02023]$
	temperature + class bias	-0.00141	-0.01679	$[-0.01814, -0.01544]$
IETrans+Rwt vs base, log	temperature	-0.42692	-0.05435	$[-0.05786, -0.05085]$
	shared temperature	-0.37726	-0.09225	$[-0.09706, -0.08745]$
	temperature + class bias	$+0.00831$	-0.06609	$[-0.06988, -0.06230]$

Table 34. **Recall of the SGCls and SGDet checkpoints**, released (repository README) and replayed by the inline evaluator (graph constraint), on the full test set.

Protocol	Model	R@50 released	R@50 replayed	mR@50 released	mR@50 replayed
SGCLS	base	0.3925	0.3924	0.0802	0.0802
	TDE	0.2631	0.2631	0.1321	0.1322
	LA_1	–	0.3840	–	0.1280
SGDET	base	0.3245	0.3247	0.0583	0.0583
	TDE	0.1656	0.1657	0.0894	0.0901
	LA_1	–	0.3186	–	0.0966

predicate it keeps the first-hit rank with and without the graph constraint, ordering pairs by the post-processor's triple score (predicate times subject and object scores) and matching a pair when both its labels equal the ground truth and both boxes overlap it at IoU ≥ 0.5, and it keeps the predicate distribution of the matched pair with the largest summed IoU, chosen from boxes and labels alone and so the same for all three models. On test images 0–199 the replayed TDE recalls equal the codebase's evaluator in both protocols to its printed precision, for R, mR, ng-R and ng-mR at $K = 20, 50, 100$; on the full test set they reproduce the released numbers (Tab. 34).

The grouping is the ground-truth subject–object class pair, as in PredCls. A relation whose objects the model fails to recognise is a miss for all three models under every predicate label, so it adds to neither part; the AUC and the proper score use the relations with a matched pair (58% in SGCls).

Table 35 gives the result. In SGCls TDE raises mR@50 by $+0.0517$, of which $+0.0510$ is group-level (99%) and $+0.0008$, interval $[-0.0024, +0.0040]$, case-level. The logit-adjusted control gains as much, $+0.0518$, at R@50 $= 0.3840$ against TDE's 0.2631, and TDE's case-level part against it is -0.0027, $[-0.0063, +0.0009]$. On relations with a matched pair TDE's within-cell discrimination falls: the AUC change weighted as the split weights cells is -0.0128, interval $[-0.0171, -0.0074]$ (-0.0047, $[-0.0122, +0.0031]$, by comparison), and the matched quadratic case-level part is -0.00540, $[-0.00655, -0.00425]$. In SGDet, on all $26{,}446$ test images, TDE raises mR@50 by $+0.0310$, 87% group-level, with a case-level part of $+0.0041$, $[+0.0013, +0.0070]$, which the control matches exactly (TDE against the control: -0.0000, $[-0.0030, +0.0030]$); the control gains more, $+0.0418$, at R@50 $= 0.3186$ against TDE's 0.1657; neither AUC weighting detects a change in TDE's within-cell ranking (-0.0012, $[-0.0051, +0.0030]$, relation-weighted), and the matched quadratic case-level part is negative, -0.00446, $[-0.00560, -0.00332]$.

Q.6. An input-grounded audit outside scene graphs: Waterbirds

Waterbirds [34] pastes landbirds and waterbirds on land and water backgrounds so that in training the background agrees with the label 95% of the time. The cell here is the background's scene category (bamboo forest, forest, lake, ocean), a property of the input, and the test split holds $5{,}794$ images. Models are logistic regressions on frozen CLIP ViT-L/14 features

Table 35. **The audit in SGCls and SGDet.** Mean recall@50 split (graph constraint, micro-averaged, grouping by ground-truth class pair) with its image-clustered case-level interval and group-level share, and the within-cell AUC change on relations with a matched pair, by comparison (c) and as the split weights cells (r); image bootstrap.

Protocol	Comparison	ΔmR	group	case	case 95% CI	group share	$\Delta\mathrm{AUC}_c$	$\Delta\mathrm{AUC}_r$
SGCLS	TDE vs base	$+.0517$	$+.0510$	$+.0008$	$[-.0024, +.0040]$	99%	$-.0047$ $[-.0122, +.0031]$	$-.0128$ $[-.0171, -.0074]$
	LA_1 vs base	$+.0518$	$+.0483$	$+.0035$	$[+.0003, +.0067]$	93%	$-.0000$ $[-.0000, +.0000]$	$-.0000$ $[-.0000, +.0000]$
	TDE vs LA_1	$-.0001$	$+.0026$	$-.0027$	$[-.0063, +.0009]$	–	$-.0047$ $[-.0122, +.0031]$	$-.0128$ $[-.0171, -.0074]$
SGDET	TDE vs base	$+.0310$	$+.0269$	$+.0041$	$[+.0013, +.0070]$	87%	$+.0011$ $[-.0049, +.0079]$	$-.0012$ $[-.0051, +.0030]$
	LA_1 vs base	$+.0418$	$+.0377$	$+.0041$	$[+.0016, +.0067]$	90%	$+.0000$ $[-.0000, +.0000]$	$+.0000$ $[-.0000, +.0000]$
	TDE vs LA_1	$-.0108$	$-.0108$	$-.0000$	$[-.0030, +.0030]$	100%	$+.0011$ $[-.0049, +.0079]$	$-.0012$ $[-.0051, +.0030]$

Table 36. **Waterbirds, GRW against ERM**: Brier split with the log-score case-level part, test split, cells by background scene; below, accuracy, worst-group accuracy and within-cell AUC.

Comparison (cells: background scene)	ΔT	group	case	case 95% CI	case (log)	95% CI
GRW vs ERM	$+0.06373$	-0.02766	$+0.09139$	$[+0.08494, +0.09785]$	$+0.22455$	$[+0.21256, +0.23653]$
ERM recalibrated vs ERM	$+0.01788$	$+0.06348$	-0.04560	$[-0.04944, -0.04177]$	-0.28306	$[-0.29469, -0.27144]$
GRW vs ERM, both recalibrated	$+0.05812$	-0.05563	$+0.11375$	$[+0.10726, +0.12025]$	$+0.34510$	$[+0.32996, +0.36023]$

Model	accuracy	worst-group acc.	within-cell AUC
ERM	0.8952	0.6433	0.9723
GRW	0.9391	0.8536	0.9754

(`colab_waterbirds_features.py`, `run_waterbirds_audit.py`): ERM, and GRW, which reweights training images so that each (label, background) group counts equally, the standard debiasing intervention and the analogue of TDE here. Temperature and a per-class bias are fitted on the group-balanced validation split and both models are evaluated on the test split.

GRW raises accuracy from 0.8952 to 0.9391 and worst-group accuracy from 0.6433 to 0.8536 (Tab. 36). Its Brier gain is $+0.06373$, and unlike TDE in PredCls it is not a change of per-background priors: the group-level part is -0.02766 and the case-level part $+0.09139$, interval $[+0.08494, +0.09785]$; with both models recalibrated the same way the parts are -0.05563 and $+0.11375$, $[+0.10726, +0.12025]$. The within-cell AUC rises only from 0.9723 to 0.9754 ($+0.0031$, $[+0.0009, +0.0057]$), so what GRW gains over ERM is mostly better-calibrated confidence in a bird's own label within a background, rather than a different ranking of the birds. Recalibration alone moves both parts (-0.04560 case-level), which is why the matched comparison recalibrates both models. Four cells are coarse, and the result holds with the two backgrounds (land, water) as cells. It is one model pair on one dataset: it shows that the split can return a case-level answer where TDE's is group-level, not that group reweighting is generally case-level.

Q.7. Zero-shot CLIP on the canonical split

The CLIP model of Sec. 6 trains a head on Visual Genome labels. As a first look at a vision–language model used with no training at all, we score the canonical test relations of Sec. 5 zero-shot: frozen CLIP ViT-B/32 [31] encodes the union box of subject and object and is compared with one prompt per predicate, "a photo of a {subject} {predicate} a {object}" (`colab_clip_zeroshot.py`). The map from dataset index to image file follows the codebase's loader, and every image's size matches the one stored with the audit. Nothing is trained, so the model has no predicate prior of its own. FREQ is the MOTIFS frequency branch, constant within a cell up to 31 relations whose scored pair is a duplicate box with another label (hence its AUC of 0.5002). Against FREQ a model's case-level part is its own within-cell information. "Zero-shot CLIP + FREQ" multiplies the two distributions and renormalises, which leaves the within-cell AUC unchanged. Protocol as in Sec. 5: per-model temperature on the calibration half, split on the other.

Zero-shot CLIP ranks the correct predicate first for 2.5% of relations, and its group-level loss against FREQ is -0.51900. It does carry within-cell information: its AUC is 0.5164, above the 0.5 of any model constant within cells, and its case-level part over FREQ is $+0.00046$, interval $[+0.00041, +0.00051]$. But this is a small fraction of what MOTIFS learns from Visual Genome: MOTIFS's AUC is 0.5705, the difference -0.0542, interval $[-0.0655, -0.0423]$, and its case-level part over FREQ is $+0.02943$. Given the training prior, zero-shot CLIP reaches 0.6289 top-1, but its case-level part over FREQ ($+0.00989$) is a third of MOTIFS's. Frozen ViT-B/32 with one prompt per predicate is a weak baseline, at 2.5% top-1 close

Table 37. **Zero-shot CLIP ViT-L/14 with a five-prompt ensemble**, as in Tab. 38.

Comparison	ΔT	group	case	case 95% CI	case (log)	95% CI
zero-shot CLIP vs FREQ	-0.51929	-0.51962	$+0.00033$	$[+0.00030, +0.00037]$	$+0.00785$	$[+0.00719, +0.00851]$
MOTIFS vs FREQ	$+0.02382$	-0.00561	$+0.02943$	$[+0.02777, +0.03108]$	$+0.11265$	$[+0.10740, +0.11789]$
zero-shot CLIP vs MOTIFS	-0.54311	-0.51401	-0.02910	$[-0.03075, -0.02744]$	-0.10480	$[-0.10986, -0.09974]$
zero-shot CLIP + FREQ vs FREQ	-0.07729	-0.08989	$+0.01260$	$[+0.01081, +0.01438]$	$+0.06150$	$[+0.05634, +0.06665]$
zero-shot CLIP + FREQ vs MOTIFS	-0.10111	-0.08428	-0.01683	$[-0.01905, -0.01461]$	-0.05115	$[-0.05713, -0.04518]$
TDE vs FREQ	-0.15882	-0.18819	$+0.02937$	$[+0.02778, +0.03096]$	$+0.15661$	$[+0.14905, +0.16418]$

Model	top-1 acc.	within-cell AUC	T^*
zero-shot CLIP	0.0179	0.5252	6.47
zero-shot CLIP + FREQ	0.6232	0.5252	0.83
FREQ	0.6802	0.5002	0.77
MOTIFS	0.6981	0.5705	2.50
TDE	0.5577	0.5731	0.92

Table 38. **Zero-shot CLIP on the canonical relations.** Upper block: matched split (quadratic, with the log-score case-level part); lower block: top-1 accuracy, within-cell AUC and fitted temperature. The AUC of zero-shot CLIP + FREQ equals zero-shot CLIP's by construction.

Comparison	ΔT	group	case	case 95% CI	case (log)	95% CI
zero-shot CLIP vs FREQ	-0.51854	-0.51900	$+0.00046$	$[+0.00041, +0.00051]$	$+0.01018$	$[+0.00915, +0.01121]$
MOTIFS vs FREQ	$+0.02382$	-0.00561	$+0.02943$	$[+0.02777, +0.03108]$	$+0.11265$	$[+0.10740, +0.11789]$
zero-shot CLIP vs MOTIFS	-0.54236	-0.51339	-0.02897	$[-0.03061, -0.02732]$	-0.10247	$[-0.10750, -0.09743]$
zero-shot CLIP + FREQ vs FREQ	-0.06544	-0.07533	$+0.00989$	$[+0.00815, +0.01162]$	$+0.04591$	$[+0.04129, +0.05053]$
zero-shot CLIP + FREQ vs MOTIFS	-0.08926	-0.06972	-0.01954	$[-0.02175, -0.01733]$	-0.06674	$[-0.07261, -0.06087]$
TDE vs FREQ	-0.15882	-0.18819	$+0.02937$	$[+0.02778, +0.03096]$	$+0.15661$	$[+0.14905, +0.16418]$

Model	top-1 acc.	within-cell AUC	T^*
zero-shot CLIP	0.0253	0.5164	3.73
zero-shot CLIP + FREQ	0.6289	0.5164	0.83
FREQ	0.6802	0.5002	0.77
MOTIFS	0.6981	0.5705	2.50
TDE	0.5577	0.5731	0.92

to chance. A larger model with prompt ensembles (below) changes the picture little, but region-level features or fine-tuning may carry more, and we draw no general conclusion about vision–language models from it. What it shows is that the split separates a model with no predicate prior and little within-pair information from one with both, in line with evidence that such models often treat captions as bags of words [48]. TDE's case-level part over FREQ, $+0.02937$, equals MOTIFS's to the third decimal, as the null of Sec. 5 implies.

A larger model. Repeating the audit with CLIP ViT-L/14 (OpenAI weights) and an ensemble of five prompt templates per predicate (`colab_clip_zeroshot.py`) gives the same picture (Tab. 37). Top-1 accuracy is 1.8%, no better than ViT-B/32's, but the within-cell AUC rises to 0.5252, still below MOTIFS's 0.5705 by -0.0453, interval $[-0.0563, -0.0307]$. Its case-level part over FREQ is $+0.00033$, $[+0.00030, +0.00037]$, as small as before; given the training prior it is $+0.01260$, $[+0.01081, +0.01438]$, against ViT-B/32's $+0.00989$ and less than half of MOTIFS's. A stronger zero-shot model gains within-pair ranking but not near the trained one.